

WHAT MATTERS FOR REPRESENTATION ALIGNMENT: GLOBAL INFORMATION OR SPATIAL STRUCTURE?

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ABSTRACT

Representation alignment (REPA) steers generative training by distilling the representations of a powerful, pretrained vision encoder into the intermediate features of a diffusion model. In this work we interrogate a foundational question that has thus far received little scrutiny: which property of the target representation actually governs generation—its *global* semantic content (as quantified, for instance, by ImageNet-1K accuracy), or its spatial structure (namely, the pairwise cosine similarity among patch tokens)? The prevailing conviction maintains that superior global semantic performance translates directly into a more effective alignment target. To adjudicate this claim, we first undertake a large-scale empirical investigation spanning 27 distinct vision encoders together with several model scales. The outcome is counterintuitive: it is spatial structure, rather than global performance, that dictates the generative utility of a target representation. Motivated by this observation, we introduce two elementary modifications whose sole purpose is to accentuate the transfer of *spatial* information—supplanting the conventional MLP projection layer of REPA with a simple convolutional layer, and appending a spatial normalization layer to the external representation. Remarkably, this minimal recipe (realized in fewer than four lines of code), which we term iREPA, yields consistent acceleration of REPA convergence across a heterogeneous collection of vision encoders, model sizes, and training regimes (including REPA, REPA-E, Meanflow, and JiT, among others). Taken together, our findings call for a reappraisal of the fundamental mechanism underlying representational alignment and of how it may be harnessed to train generative models more effectively.

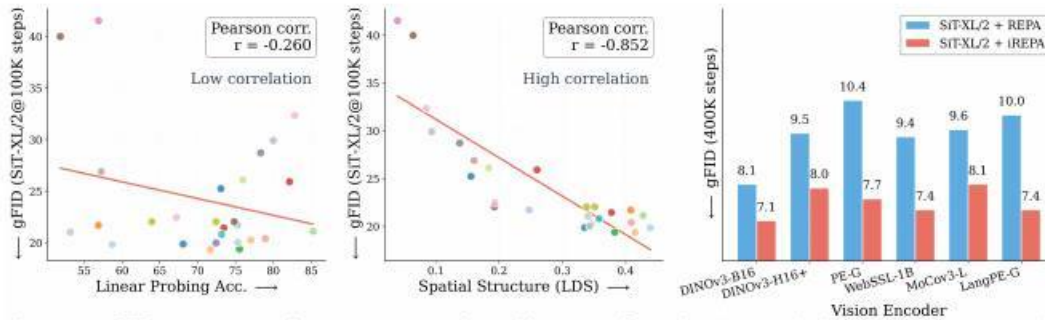


Figure 1: **What matters for representation alignment?** *Left:* A correlation analysis across 27 diverse vision encoders. Strikingly, and contrary to prevailing wisdom, we find that it is spatial structure—rather than global performance (as measured by linear probing accuracy)—that governs the generation performance of a target representation. *Right:* We interrogate this finding further by introducing two simple modifications that accentuate the transfer of spatial features from the target representation to the diffusion model. Our minimal approach consistently accelerates REPA convergence across diverse settings.



Figure 2: **Motivating examples — spatial structure matters.** A comparison of metrics that lays bare the inverse relationship between ImageNet accuracy and generation quality. *Left:* PE-G, its significantly higher validation accuracy (82.8% vs. 53.1%) notwithstanding, exhibits worse generation quality than SpatialPE-B (Bolya et al., 2025). *Right:* In like manner, WebSSL-1B (Fan et al., 2025) attains much stronger global performance (76.0% vs. 53.1%) yet weaker generation. **Spatial Self-Similarity:** We find that spatial structure instead furnishes a more reliable predictor of generation quality than does global performance. See §3 for the spatial structure metric. All results are reported at 100K using SiT-XL/2 and REPA.

1 INTRODUCTION

Representation alignment has lately emerged as a potent instrument for accelerating the training of diffusion transformers (Ma et al., 2024; Peebles & Xie, 2023). By reconciling the internal representations of a diffusion model with those of pretrained self-supervised visual encoders, recent methods have attained pronounced gains in both convergence speed and terminal performance (Yu et al., 2024; Leng et al., 2025a). Yet, notwithstanding these empirical successes, our understanding of the precise mechanisms by which self-supervised features augment diffusion model training remains conspicuously incomplete. A foundational question thus persists: is the observed improvement principally attributable to the incorporation of richer global semantic information—as is customarily gauged through linear probing performance—or does it instead arise from a more faithful preservation of spatial structure, that is, the relational geometry among patch token representations?

Elucidating these mechanisms is of considerable consequence for the advancement of generative model training, since it bears directly upon one’s capacity to select an optimal target representation and to extract its full benefit. At present, a widespread assumption holds that an encoder’s efficacy for representation alignment is tightly coupled to its ImageNet-1K validation accuracy, itself a proxy for global semantic understanding (Oquab et al., 2024; Chen et al., 2021). On this view, target representations exhibiting superior ImageNet performance are conjectured to beget superior generation (Yu et al., 2024). Accordingly, improvements in the linear probing accuracy of diffusion features are routinely adduced as evidence for the efficacy of representation alignment, thereby entrenching the primacy accorded to global semantic information as the principal engine of improvement. The following passage from REPA (Yu et al., 2024) crystallizes the prevailing understanding:

“When a diffusion transformer is aligned with a pretrained encoder that offers more semantically meaningful representations (i.e., better linear probing results), the model not only captures better semantics but also exhibits enhanced generation performance, as reflected by improved validation accuracy with linear probing and lower FID scores.”

In this paper, we contest this conventional wisdom, subjecting to systematic scrutiny the question of what genuinely drives representation alignment: global semantic information or spatial structure? Through a large-scale empirical analysis encompassing a diverse array of vision encoders—among them recent large-scale vision foundation models such as WebSSL (Fan et al., 2025), DINOv3 (Siméoni et al., 2025), perceptual encoders (Bolya et al., 2025), and C-RADIO (Heinrich et al., 2024)—we arrive at three surprising findings.

Higher validation accuracy does not imply better representation for generation. Contrary to prevailing assumptions, vision encoders endowed with superior global semantic performance—as measured by ImageNet-1K linear probing accuracy—can frequently prove inferior for generation. Consider, by way of illustration, PE-Spatial-B, a spatially-tuned model derived from PE-Core-G (Bolya et al., 2025). We observe that although PE-Spatial-B attains a substantially lower validation

accuracy¹ on patch tokens than PE-Core-G (53.1% vs. 82.8%), it nonetheless yields superior generation under REPA (Figure 2). In the same vein, WebSSL-1B (Fan et al., 2025) exhibits markedly stronger global performance (76.0% vs. 53.1%), yet weaker generation. In Section §2 we establish that this pattern recurs across a wide range of target representations, intimating a fundamental principle governing how representation alignment confers benefit upon diffusion training.

Spatial structure not global information determines generation performance. To render this insight quantitative, we introduce several elementary metrics that characterize the spatial self-similarity structure (Shechtman & Irani, 2007) among patch tokens (§3). We subsequently conduct a large-scale analysis, computing the correlation of REPA generation FID with linear probing accuracy and with spatial structure across 27 vision encoders and 3 model scales (SiT-B, SiT-L, SiT-XL). Strikingly, every spatial structure metric correlates far more strongly (Pearson $|r| > 0.852$) with generation FID than does ImageNet-1K validation accuracy ($|r| = 0.26$), whose predictive power is comparatively negligible. These findings are further corroborated by supplementary experiments demonstrating that external representations possessing only meager global information can nonetheless deliver substantial gains under REPA. SAM2, for instance, engenders better REPA generation than encoders whose ImageNet accuracy is roughly 60% higher (Fig. 3). Likewise—though admittedly not optimal—we find that classical spatial descriptors such as SIFT (Lowe, 1999) and HOG (Dalal & Triggs, 2005) can also be marshaled to secure respectable gains through representation alignment (§3).

Accentuating spatial features improves convergence performance. To probe this further, we introduce two elementary modifications (fewer than four lines of code) that expressly amplify the transfer of “spatial” information from the target representation to the diffusion model: (1) we first introduce a spatial regularization layer that heightens the spatial contrast of the target representations; and (2) we then demonstrate that the standard MLP-based projection layer—employed to map diffusion features onto the dimensionality of the target representation—incurs a loss of spatial information (Figure 6a), and we forestall this loss by substituting a simple convolutional layer. The results are striking: our minimal method, termed iREPA, consistently accelerates REPA convergence across diverse variations in encoders, model sizes, and training recipes, including REPA-E (Leng et al., 2025a) and Meanflow (Geng et al., 2025) with REPA.

We summarize the principal contributions of this paper as follows:

- We conduct a large-scale empirical analysis establishing that it is spatial structure, rather than global semantic information, that governs the efficacy of representation alignment.
- We introduce a straightforward and inexpensively computed Spatial Structure Metric (SSM), which exhibits substantially higher correlation with downstream FID performance than linear probing scores.
- We propose simple modifications that more effectively accentuate the transfer of spatial information from the target representation to diffusion features. Our minimal method delivers consistent improvements in convergence speed over REPA across variations in target representation, model architecture, and training recipe (REPA, REPA-E, Meanflow with REPA, JiT with REPA, and others).

2 MOTIVATION: GLOBAL INFORMATION MATTERS LESS

We begin by adducing several motivating examples which demonstrate that a target representation of higher global performance (ImageNet-1K accuracy) does not entail better generation performance under REPA. We subsequently show in §3 that these hitherto-unexplained observations admit a far more cogent account once the spatial structure of the target representation is brought into consideration.

Recent vision encoders. We examine a selection of recent vision encoders, including Perceptual Encoders (Bolya et al., 2025), WebSSL (Fan et al., 2025), and DINOv3 (Siméoni et al., 2025). Consider PE-Spatial-B (80M), a compact spatially-tuned model derived from PE-Core-G (1.88B) (Bolya et al., 2025). As Figure 2 attests, although PE-Core-G attains a considerably higher ImageNet-1K accuracy (82.8% vs 53.1%), it fares worse when deployed as the target representation for REPA (FID 32.3 vs 21.0). An analogous reversal holds for WebSSL-1B (1.2B), which, despite its

¹Following REPA (Yu et al., 2024), we employ linear probing accuracy on patch tokens as our measure of the global semantic performance of an external representation, since only patch tokens participate in representation alignment.

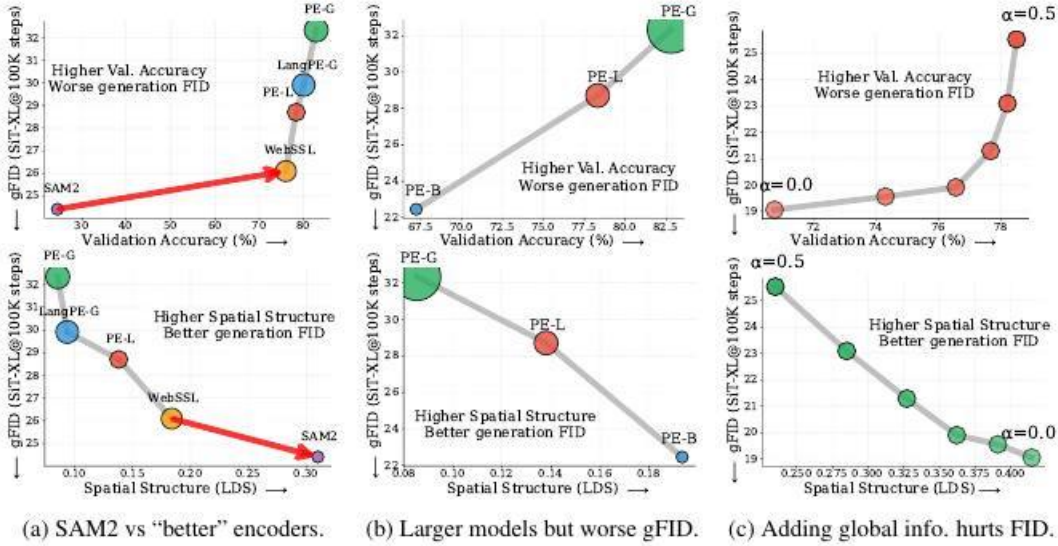


Figure 3: **Higher global information does not imply better REPA performance.** **Top row:** Several trends attesting that global performance correlates poorly with generation FID under REPA. (a) SAM2-S, with a validation accuracy of only 24.7%, nonetheless yields better generation performance under REPA than other models whose validation accuracy is roughly 60% higher. (b) Larger encoders within a given family may enjoy better validation yet deliver worse generation performance. (c) Injecting global information into patch tokens via the CLS token improves global performance but impairs generation. **Bottom row:** We demonstrate that spatial structure, rather than global performance, constitutes a more reliable indicator for generation. Please see §3 for a detailed large-scale analysis across the several spatial structure metrics. All results are reported w/o classifier free guidance, SiT-XL/2 w/ REPA and 250 NFE (Yu et al., 2024) for inference.

markedly higher ImageNet-1K accuracy (76.0% vs 53.1%), likewise underperforms as a REPA target representation (FID 26.1 vs 21.0).

SAM2 outperforms vision encoders with much higher ImageNet-1K accuracy. To further illuminate how little global information bears upon generation performance, we scrutinize the SAM2-S vision encoder (46M) (Ravi et al., 2024)—a diminutive model harboring very little global information, with a validation accuracy of merely 24.1% (Fig. 3a). Surprisingly, when enlisted for REPA, SAM2-S attains a better FID than other vision encoders of substantially higher ImageNet-1K accuracy, PE-Core-G (82.8%) among them.

Larger models within same encoder family can have similar or worse generation performance. A commonly held perception is that larger models within a given encoder family possess superior representations (as measured by ImageNet-1K accuracy). For representation alignment, however, the larger variants often precipitate comparable (DINOv2) or even *degraded* (PE, Cradio) generation performance (Fig. 3b). Notably, (Yu et al., 2024) report a kindred observation for DINOv2 and attribute it to the conjecture that “all DINOv2 models being distilled from the DINOv2-g model and thus sharing similar representations”. We later demonstrate that this trend, too, is better accounted for through spatial structure (§3).

Adding more global information can hurt generation. To ascertain whether additional global information benefits representation alignment, we perform controlled experiments that inject global semantics—via the CLS token—into the local patch tokens (with DINOv2 as encoder). The CLS token mixing operation is given by $\mathbf{p}_i^{\text{new}} = \mathbf{p}_i + \alpha \cdot \mathbf{c}$, wherein $\mathbf{p}_i$ denotes the i^{th} patch token, $\mathbf{c}$ the CLS token, and $\alpha \in [0, 0.5]$ modulates the mixing strength. As Figure 3c illustrates, as α ascends from 0 to 0.5, linear probing accuracy improves monotonically from 70.7% to 78.5%. Generation quality, however, deteriorates markedly in tandem, with FID degrading from 19.2 at $\alpha = 0$ to 25.4 at $\alpha = 0.5$.

The foregoing observations collectively underscore that the global performance of a representation constitutes a poor indicator of its REPA performance. We show next (§3) that spatial structure instead furnishes a far more reliable signal.

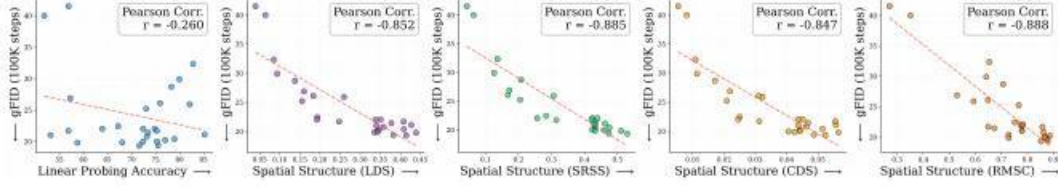


Figure 4: **Spatial structure exhibits higher correlation with generation quality than does linear probing.** A correlation analysis across 27 diverse vision encoders, with SiT-XL/2 and REPA. Linear probing correlates only weakly with FID (Pearson $|r| = 0.260$), whereas every spatial structure metric—LDS ($|r| = 0.852$), SRSS ($|r| = 0.885$), CDS ($|r| = 0.847$), and RMSC ($|r| = 0.888$)—displays a considerably stronger correlation with generation performance. See Fig. 10 for detailed plots annotated with encoder labels.

- **Finding 1.** Superior global semantic information (e.g., ImageNet-1K accuracy) does not entail a superior representation for generation.

3 SPATIAL STRUCTURE MATTERS MORE

We advance the hypothesis that spatial structure, rather than global information, is what dictates the generation performance attainable with a target representation. To render this hypothesis testable, we first devise several elementary metrics for quantifying the spatial self-similarity structure (Shechtman & Irani, 2007) of target representations. We then demonstrate that all such spatial structure metrics not only correlate considerably more strongly with gFID than does ImageNet-1K accuracy, but also serve to explain the previously unaccounted-for observations of §2.

Measuring spatial self-similarity structure. Given an image $\mathcal{I}$ and vision encoder $\mathcal{E}$, we define:

- *Self-similarity.* Let $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ be the extract patch representations, with $T = H \times W$ patches. Let kernel $K_{\mathcal{X}}(\cdot, \cdot) \in \mathbb{R}$ measure appearance self-similarity (Shechtman & Irani, 2007) between patch tokens. Here, we use the cosine kernel $K_{\mathcal{X}}(t, t') = \frac{(\mathbf{x}_t, \mathbf{x}_{t'})}{\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2}$.
- *Spatial distance.* Let $\mathcal{P} \in \mathbb{N}^{T \times 2}$ be the spatial location of each of the T tokens in coordinate space and $d(\cdot, \cdot) \in \mathbb{N}$ be the Manhattan distance between pairs of tokens.

We then define the *spatial* self-similarity metric in terms of the manner in which self-similarity $K_{\mathcal{X}}$ varies with the lattice distance d separating patch tokens. Intuitively, larger values betoken stronger spatial organization, whereby proximate patches are more mutually similar than distant ones. By default, we adopt a simple correlogram-contrast (local vs. distant) metric (Huang et al., 1997):

$$\text{LDS}(\mathcal{X}, \mathcal{P}) = \mathbb{E}_{d(t, t') \in (0, r_{\text{near}})} K_{\mathcal{X}}(t, t') - \mathbb{E}_{d(t, t') \geq r_{\text{far}}} K_{\mathcal{X}}(t, t').$$

The final spatial self-similarity score (LDS) is obtained as the expectation over the patch representation $\mathcal{X}$. Here we set $r_{\text{near}} = r_{\text{far}} = H/2$, though we find the correlation to be robust to the precise choice of these radii. We refer the reader to Appendix B for exact details, as well as for alternative spatial metrics (CDS, SRSS, RMSC) that likewise prove effective (see Fig. 4).

Spatial structure correlates much higher with generation performance than linear probing. We next undertake a large-scale correlation analysis spanning 27 diverse vision encoders. As Figure 4 shows, whereas the customarily employed linear probing exhibits only a very weak correlation (Pearson $|r| = 0.26$), every SSM metric displays a considerably stronger correlation with generation performance (Pearson $|r| > 0.85$).

Generalization across model scales. We further corroborate this correlation across model scales (SiT-B, SiT-L, SiT-XL) in Figure 5. Linear probing manifests a weak correlation throughout ($|r| < 0.306$), while spatial structure retains a markedly stronger correlation with generation performance ($|r| > 0.826$).

- **Finding 2.** Spatial structure correlates far more strongly with generation performance than does linear probing. We proceed to invoke the spatial metrics to explain hitherto-unexplained trends.

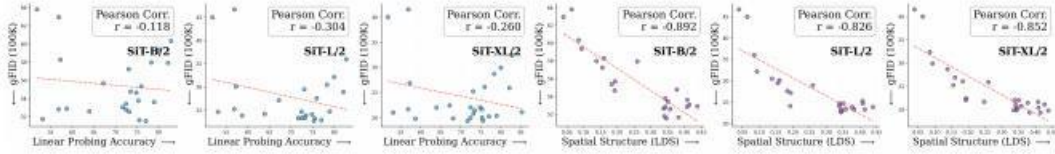


Figure 5: **Correlation analysis across model scales.** Across the several model scales, we find that spatial structure (right) consistently exhibits a stronger correlation with gFID than does linear probing (left).

Spatial structure can explain previously unexplained trends. As discussed in §2, global performance (ImageNet-1K accuracy) does not constitute a predictive measure of efficacy for representation alignment. We find, in contradistinction, that the spatial metrics introduced above serve as considerably more reliable predictors.

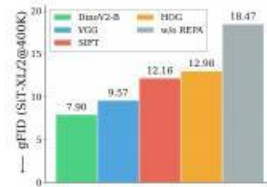
(1) *PE-Spatial-B vs PE-Core-G*: Figure 2 shows that PE-Core-G attains a much higher ImageNet-1K accuracy (82.8% vs 53.1%) yet fares worse as a target representation for REPA (FID 32.3 vs 22.0). Recourse to the spatial structure metric clarifies the matter: as Figure 2 reveals, PE-Spatial-B—its lower global performance notwithstanding—exhibits appreciably better spatial structure than PE-Core-G, which duly accounts for the superior generation performance observed.

(2) *SAM2 outperforms “better” vision encoders*: §2 establishes that although SAM2 attains a much lower ImageNet-1K accuracy (24.1%), it engenders better generation than encoders whose accuracy is roughly 60% higher. As shown in Fig. 3a, these gains are directly explicable in terms of SAM2’s superior spatial structure.

(3) *Larger models in same encoder family underperform*: As Fig. 3b shows, although larger models within a given encoder family enjoy higher ImageNet-1K accuracy, they may possess inferior spatial structure, and consequently yield worse generation performance under REPA. This concurs with the observations of Yu et al. (2024), who likewise find that larger models can deliver comparable or worse generation performance.

(4) *Adding global information to patch tokens via CLS token hurts generation*: In Figure 3c we observe that augmenting global information by mixing the CLS token into the patch tokens improves global performance. Such mixing, however, attenuates the spatial contrast among patch tokens, whereupon a patch token comes to exhibit high similarity with otherwise unrelated tokens (*e.g.*, from a foreground object to the background). This diminished spatial structure, in turn, translates into worse generation performance.

If spatial structure matters more, can we use SIFT or HOG features for REPA? Surprisingly, the answer is affirmative. We find that, though certainly not optimal, classical spatial descriptors such as SIFT (Lowe, 1999), HOG (Dalal & Triggs, 2005), and intermediate VGG features (Simonyan & Zisserman, 2014) all confer performance gains under REPA. This constitutes further evidence that representation alignment can profit from spatial features in isolation, without recourse to any additional global information.



Can we use spatial metrics to explain gains with REPA? Indeed we can. As demonstrated, the spatial metrics introduced here serve to explain both the gains realized with REPA and those of our improved training recipe (iREPA), which we present next in §4.



4 iREPA: IMPROVING REPRESENTATION ALIGNMENT BY ACCENTUATING WHAT MATTERS

To interrogate more closely the role of spatial structure in representation alignment, we introduce two elementary modifications to the original REPA training recipe, each designed to enhance the transfer of spatial features from the teacher (the vision encoder) to the student (the diffusion transformer).

Convolutional projection layer instead of MLP. The standard REPA formulation employs a 3-layer MLP projection to map the dimensionality of the diffusion features onto that of the external

representation. As Figure 6a reveals, however, this projection proves lossy, attenuating the spatial contrast among patch tokens. We therefore supplant the MLP with a lightweight convolutional layer (kernel size 3, padding 1) that operates directly upon the spatial grid. By virtue of its inductive bias, the convolutional structure naturally preserves local spatial relationships.

Spatial normalization layer. In accord with Siméoni et al. (2025), we find that the patch tokens of pretrained vision encoders harbor a substantial global component—a fact evidenced by the high linear probing scores attained by the mean of patch tokens (Figure 14). Moreover, Figure 6b makes plain that, while mixing this global information (the mean of patch tokens) into the patch tokens improves global performance, it does so at the expense of the spatial contrast among individual patch tokens, whereby a token (e.g., a foreground object) comes to exhibit high similarity with otherwise unrelated tokens (e.g., the background).

In light of the results of §3, we hypothesize that this global information (the mean of patch tokens) may be relinquished in order to heighten the spatial contrast among patch tokens. The resulting improvement in contrast ought to furnish a stronger spatial signal (the pairwise similarity between patch tokens), and thereby better REPA performance. To this end, we append a simple spatial normalization layer (Ulyanov et al., 2016) to the patch tokens of the target representation:

$$\mathbf{y} = \frac{\mathbf{x} - \gamma \mathbb{E}[\mathbf{x}]}{\sqrt{\text{Var}[\mathbf{x}] + \epsilon}},$$

where $\mathbf{x} \in \mathbb{R}^{B \times T \times D}$ denotes the patch tokens, the expectation and variance are taken over the spatial dimension, and $\epsilon = 10^{-6}$ is included for numerical stability.

Algorithm 1 Summary of key iREPA modifications to standard REPA training

```
# 1. Conv projection instead of MLP
proj_layer = nn.Conv2d(D_in, D_out, kernel_size=3, padding=1)

# 2. Spatial normalization on encoder features [B, T, D]
x = x - gamma * x.mean(dim=1, keepdim=True)
x = x / (x.std(dim=1, keepdim=True) + 1e-6)
```

4.1 EXPERIMENTS

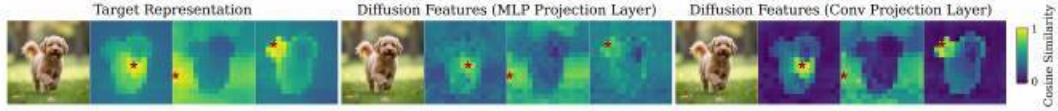
In this section we validate the improved training recipe through extensive experimentation on ImageNet 256×256 . In particular, we address the following research questions:

- Does iREPA consistently accelerate the convergence of diffusion transformers relative to REPA across a diverse range of external representations? (Figure 7, 8, 12, 13, 15)
- Is iREPA scalable with respect to model size, and does it generalize across variations in the training configuration? (Table 1 [a,b,c], 2, 4, 5, 6, and Figure 12, 16)
- Does iREPA generalize to more recent representation alignment methods, such as REPA-E (Leng et al., 2025a) and MeanFlow with REPA (Geng et al., 2025), as well as to pixel-space diffusion models such as JiT with REPA (Li & He, 2025b)? (Table 3 [a,b])

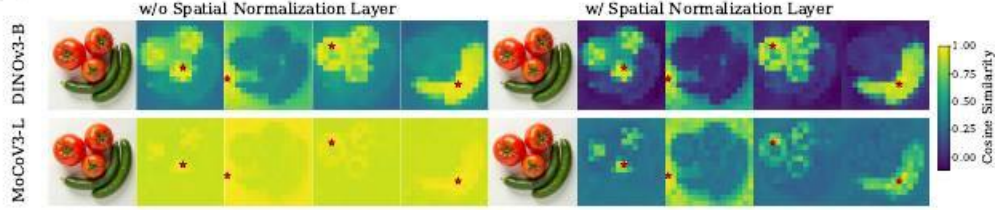
Convergence Speed. We assess the convergence behavior of iREPA across a diverse assortment of vision encoders (DINOv3-B, WebSSL-1B, PE-Core-G, CLIP-L, MoCov3, PE-Lang-G) and model sizes (SiT-XL/2 and SiT-B/2), with results reported in Fig. 7. We find that iREPA consistently secures faster convergence than baseline REPA across variations in both the target representation and the model size.

Target representation. We examine the generalization of iREPA across different vision encoders in Fig. 8 and Table 4. We observe that iREPA consistently enhances generation quality across the entire set of vision encoders. Additional comparisons spanning all 27 encoders are furnished in Appendix C, E.

Encoder size. We investigate the generalization of iREPA across encoders of differing size. Table 1a reports this analysis for PE-B (90M), PE-L (320M), and PE-G (1.88B). We find that iREPA consistently improves performance across all encoder sizes; intriguingly, the relative improvement itself grows with encoder size (22.2% for PE-B, 38.8% for PE-L, and 39.6% for PE-G).



(a) *Simpler projection layer for REPA.* The standard MLP projection layer of REPA (middle) forfeits spatial information in the course of transferring features from the target representation (left) to the diffusion features. Substituting a simpler convolutional layer, by contrast, effects a more faithful transfer of spatial information (right).



(b) *Spatial normalization layer.* The patch tokens of pretrained vision encoders harbor a global component that curtails spatial contrast, whereby the tokens of one semantic region (e.g., a tomato) come to exhibit appreciable cosine similarity with unrelated tokens (e.g., the background or a cucumber). We hypothesize that this global information may be relinquished so as to heighten the spatial contrast among patch tokens, and thereby to secure better generation performance.



(c) *Overall impact of the improved training recipe (iREPA) upon the spatial structure of the diffusion features.* Figure 6: Motivating two elementary modifications (iREPA) that enhance the transfer of spatial features from the target representation to the diffusion features. All results are reported with SiT-L/2 and REPA at 100K.

Enc. Size	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Pr. $\uparrow$	Rc. $\uparrow$	Model Size	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Pr. $\uparrow$	Rc. $\uparrow$	Aln. Depth	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Pr. $\uparrow$	Rc. $\uparrow$
PE-B (90M)	59.6	22.5	6.23	0.63	0.60	SiT-B	27.5	49.50	7.00	0.46	0.59	Layer 4	28.1	49.0	6.83	0.46	0.60
+iREPA	73.3	17.5	6.04	0.66	0.62	+iREPA	34.1	43.37	7.87	0.50	0.60	+iREPA	41.7	36.0	6.61	0.52	0.62
PE-L (320M)	48.6	28.7	6.53	0.59	0.60	SiT-L	55.7	24.10	6.25	0.62	0.60	Layer 6	26.9	50.6	7.00	0.46	0.59
+iREPA	76.3	17.6	6.36	0.65	0.61	+iREPA	66.9	20.28	6.14	0.63	0.62	+iREPA	42.4	35.3	6.90	0.53	0.62
PE-G (1.88B)	42.7	32.3	6.70	0.57	0.59	SiT-XL	70.3	19.06	5.83	0.65	0.61	Layer 8	24.7	54.8	7.35	0.44	0.58
+iREPA	70.8	19.5	6.35	0.64	0.61	+iREPA	77.9	16.96	6.26	0.66	0.61	+iREPA	38.3	38.9	7.19	0.51	0.62

(a) Vision Encoder size

(b) Model size

(c) Encoder depth (SiT-B/2)

Table 1: **Variation in training settings.** We find that iREPA delivers consistent gains over baseline REPA across a diverse array of training settings. Unless otherwise specified, all results are reported using DINOv2 as the encoder with SiT-XL/2 at 100K steps and vanilla REPA as the baseline.

Scalability. We assess the scalability of iREPA across model scales in Table 1b. We observe that the spatial improvements not only yield consistent performance gains but also confer progressively larger relative gains upon larger models, thereby demonstrating that the spatial improvements scale favorably with model size.

Encoder depth. Table 1c examines the generalization of iREPA across alignment depths. All results employ SiT-B/2 at 100K iterations with DINOv3-B as the target representation. We observe consistent improvements over baseline REPA at every alignment depth considered.

Abalation on different components. We further dissect the contribution of the individual components of iREPA in Table 2. We observe that both the spatial normalization layer and the convolutional projection layer substantially improve generation quality over baseline REPA, with the best results obtained when the two are employed in conjunction.

Training recipe. Finally, we examine the generalization of iREPA across different training recipes, namely REPA-E (Leng et al., 2025a) and MeanFlow with REPA (Geng et al., 2025). Table 3 demonstrates that the spatial improvements afforded by iREPA translate into convergence gains under each of these recipes.

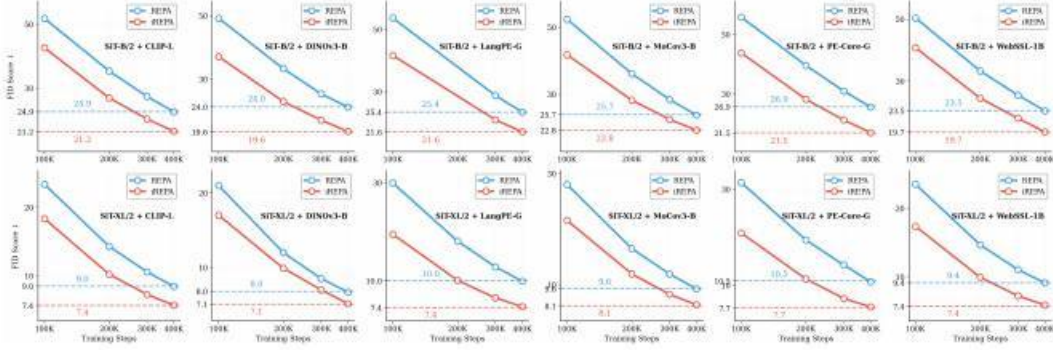


Figure 7: Accelerating spatial features consistently accelerates convergence.

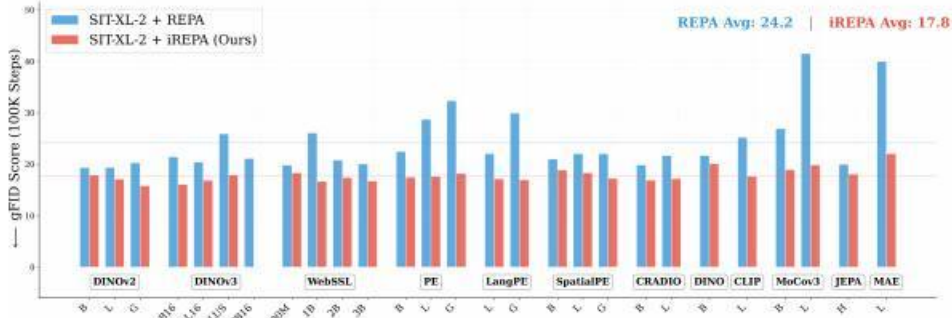


Figure 8: **Variation in target representation.** Across all 27 vision encoders, we find that accentuating the transfer of spatial features from the target representation to the diffusion features (iREPA) yields consistent improvements in convergence speed. We refer the reader to Appendix C for further results across diverse settings.

Classifier-free guidance. We evaluate the generation quality of iREPA under CFG in Table 4, which presents results across different vision encoders at 400K training iterations. We find that, across encoders, iREPA delivers faster convergence both with and without classifier-free guidance.

Pixel-space diffusion (JiT). We further evaluate iREPA on pixel-space diffusion models such as JiT-B (Li & He, 2025b). Figure 9 juxtaposes REPA and iREPA using JiT (Li & He, 2025b). We observe that accentuating the transfer of spatial information consistently attains faster REPA convergence across a variety of vision encoders (e.g., DiNOv2, DiNOv3, PE, and others).

- **Finding 3.** Accentuating the transfer of spatial structure from the target representation to the diffusion features serves to boost the convergence speed of representation alignment (REPA).

5 RELATED WORK

We canvass the most pertinent related work here and defer a more thorough discussion to Appendix H.

Representation alignment for generation. A considerable body of recent work has explored the use of external representations to improve diffusion model training (Pernias et al., 2023; Fuest et al., 2024). Notably, a series of studies (Yu et al., 2024; Yao & Wang, 2025; Leng et al., 2025a;b; Kouzelis et al., 2025) have shown that substantial performance gains accrue from aligning internal diffusion features with the clean image features of a pretrained vision encoder. This idea has been extended to video generation and 3D generation by (Zhang et al., 2025; Wu et al., 2025), respectively, while (Ma et al., 2025a) demonstrates that representation alignment can also improve the training of unified models. Despite these empirical successes, our understanding of the precise mechanisms by which

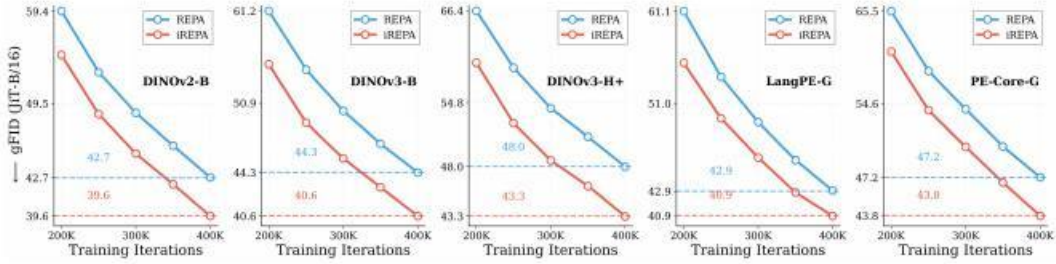


Figure 9: **Convergence comparison with pixel-space diffusion (JiT).** FID convergence curves juxtaposing REPA against iREPA with pixel-space JiT (Li & He, 2025b) across a range of vision encoders. For pixel-space diffusion models, accentuating the transfer of spatial structure consistently accelerates convergence across the several vision encoders. All results are reported with JiT-B/16 (Li & He, 2025b), a batch size of 256, and without classifier-free guidance. We refer the reader to Table 10 for further results.

Method	DINOv2-B			DINOv3-B			WebSSL-1B			PE-Core-G		
	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓
Baseline REPA	19.06	70.3	5.83	21.47	63.4	6.19	26.10	53.0	6.90	32.35	42.7	6.70
iREPA (w/o spatial norm)	18.52	73.3	6.11	17.76	74.7	5.81	21.17	64.6	6.27	24.97	57.4	6.21
iREPA (w/o conv proj)	17.66	72.8	6.03	18.28	70.8	6.18	18.44	71.0	6.22	21.72	61.5	6.26
iREPA (full)	16.96	77.9	6.26	16.26	78.8	6.14	16.66	77.5	6.18	18.19	75.0	6.03

Table 2: **Ablation on different components.** We observe that, across encoders, both the spatial normalization layer and the convolutional projection layer (§4) yield significant gains in convergence speed, with the best results obtained when the two are combined. All results are reported at 100K steps using SiT-XL/2.

self-supervised features enhance diffusion model training remains limited. In this paper, we seek to discern which aspect of the target representation actually matters for generation, and we leverage the resulting insight to propose an improved training recipe.

Spatial vs global information tradeoff in pretrained vision encoders. A number of recent works interrogate the tradeoff between global and spatial information in pretrained vision encoders (Bolya et al., 2025; Siméoni et al., 2025). (Siméoni et al., 2025) show that the continued training of self-supervised vision representations can heighten the similarity between the global CLS token and the patch tokens, thereby degrading performance on dense spatial tasks. (Bolya et al., 2025; Heinrich et al., 2024) deliberately train spatially-tuned models for dense spatial tasks. In this paper, we establish that, for generation, the spatial structure of a vision encoder matters more than its global information. We hope this will spur future research on the more judicious selection and training of external representations for generation.

6 CONCLUSION

In this paper, we have investigated what truly drives the efficacy of representation alignment: the global information or the spatial structure of the target representation? Through a large-scale empirical analysis, we arrive at a surprising conclusion: it is spatial structure, and not global information, that governs the efficacy of representation alignment. We substantiate this conclusion by introducing two simple modifications that accentuate the transfer of spatial information from the target representation to the diffusion features. Our minimal method, termed iREPA, consistently accelerates REPA convergence across diverse variations in vision encoders and training recipes. We hope our work will impel future research to reconsider the fundamental mechanism underlying representational alignment, and to determine how it may be more effectively harnessed for the training of generative models.

Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
WebSSL-1B	52.8	26.5	5.20	0.620	0.585
+iREPA-E	87.0	13.2	5.28	0.699	0.598
PE-G	50.9	25.9	5.68	0.612	0.576
+iREPA-E	80.0	16.4	5.40	0.667	0.616
DINOv3-B	82.2	14.4	4.68	0.694	0.596
+iREPA-E	93.6	11.7	4.57	0.703	0.613
DINOv2-B	87.5	12.9	5.40	0.708	0.586
+iREPA-E	91.3	12.1	4.86	0.712	0.602

(a) REPA-E (SiT-XL/2)

Encoder	w/o CFG				w/ CFG			
	4 NFE		1 NFE		4 NFE		1 NFE	
	IS $\uparrow$	FID $\downarrow$	IS $\uparrow$	FID $\downarrow$	IS $\uparrow$	FID $\downarrow$	IS $\uparrow$	FID $\downarrow$
WebSSL-1B	27.22	51.40	24.14	58.67	87.85	16.59	69.06	23.74
+iREPA	31.48	45.67	27.33	55.70	100.74	13.89	78.67	20.69
DINOv3-B	28.36	49.58	25.47	57.01	93.30	15.56	72.41	22.55
+iREPA	33.63	44.52	29.67	53.75	124.54	11.05	98.93	17.32
DINOv2-B	28.77	48.87	25.43	56.63	94.40	15.28	71.95	22.17
+iREPA	33.63	44.52	29.67	53.75	111.44	12.59	90.25	18.62

(b) MeanFlow w/ REPA (SiT-B/2)

Table 3: **Variation in training recipes.** We superimpose our spatial improvements upon REPA-E (a) and Meanflow with REPA (b), thereby securing consistent gains in both cases. For Meanflow, we report results at 1 and 4 NFEs, adopting a CFG value of 2.0. All results are reported at 100K training iterations.

Vision Encoder	Steps	w/o CFG					w/ CFG				
		IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
DINOv2-B	100K	69.20	19.3	5.89	0.64	0.61	157.2	6.35	5.91	0.769	0.536
+iREPA	100K	77.92	16.9	6.26	0.66	0.61	179.3	5.15	6.23	0.783	0.544
DINOv2-B	400K	127.4	7.76	5.06	0.70	0.66	263.0	1.98	4.60	0.799	0.610
+iREPA	400K	128.6	7.52	4.89	0.71	0.65	268.8	1.93	4.59	0.799	0.600
DINOv3-B	100K	63.64	21.4	6.14	0.63	0.60	144.0	7.57	6.09	0.762	0.526
+iREPA	100K	78.79	16.2	6.14	0.66	0.61	181.9	4.87	6.10	0.780	0.547
DINOv3-B	400K	126.7	8.10	5.06	0.70	0.66	261.2	1.99	4.58	0.799	0.609
+iREPA	400K	132.9	7.13	4.93	0.71	0.66	272.4	1.89	4.58	0.799	0.600
WebSSL-1B	100K	53.87	25.5	6.57	0.61	0.59	124.0	9.59	6.37	0.756	0.515
+iREPA	100K	77.47	16.6	6.18	0.66	0.61	177.6	5.09	6.11	0.787	0.538
WebSSL-1B	400K	116.9	9.39	5.14	0.70	0.64	250.5	2.24	4.61	0.809	0.580
+iREPA	400K	130.8	7.48	4.91	0.70	0.65	271.6	1.90	4.58	0.798	0.609
PE-Core-G	100K	42.74	32.3	6.70	0.57	0.59	97.2	14.1	6.56	0.714	0.525
+iREPA	100K	75.01	18.1	6.03	0.64	0.61	176.8	5.66	6.08	0.771	0.544
PE-Core-G	400K	109.4	10.4	5.00	0.69	0.64	238.4	2.44	4.57	0.805	0.585
+iREPA	400K	132.0	7.78	5.02	0.70	0.65	275.4	1.93	4.59	0.796	0.606

Table 4: **Results across different encoders with and w/o classifier-free guidance.** SiT-XL/2 serves as the base model. We refer the reader to Appendix E for detailed results on SiT-B/2 & SiT-L/2 across all encoders.

REPRODUCIBILITY STATEMENT

We furnish all implementation details and hyperparameters in Appendix G. Our code, model checkpoints, and analysis results will be made publicly available upon acceptance.

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A DETAILED CORRELATION ANALYSIS

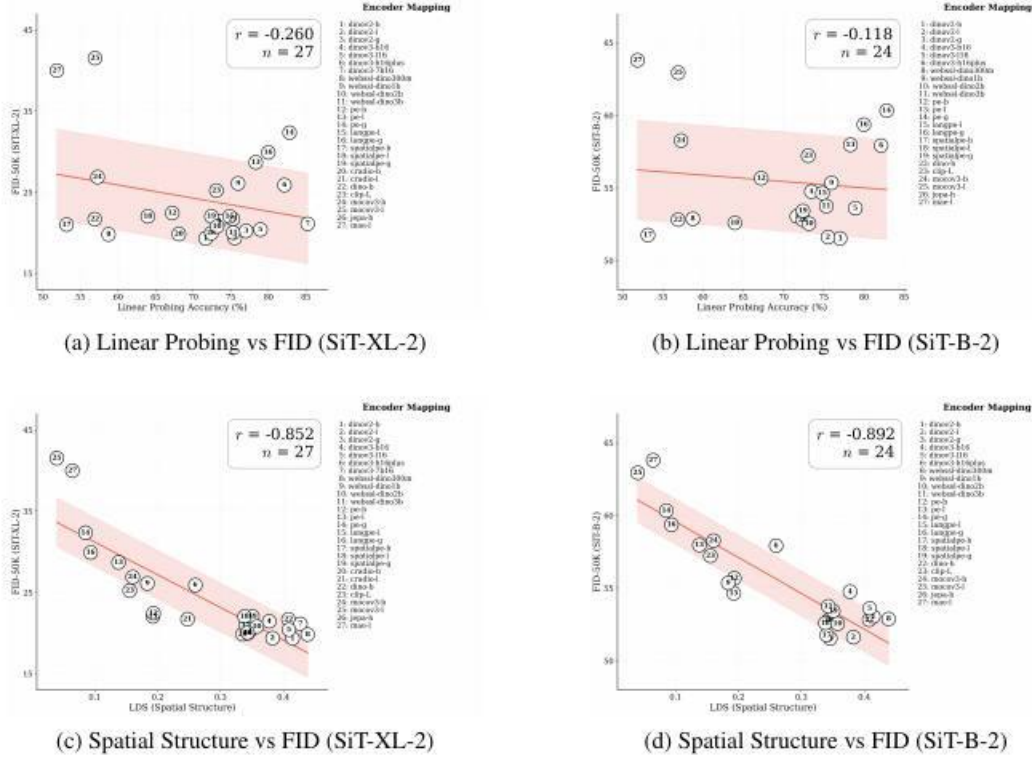


Figure 10: Detailed correlation analysis with encoder labels. The top row displays the correlation of linear probing accuracy with FID for the SiT-XL-2 and SiT-B-2 models, while the bottom row displays the correlation of spatial structure (LDS) with FID. Each point is annotated with its corresponding encoder number (see legend). The spatial structure metric evinces a significantly stronger correlation with generation quality across both model scales.

B SPATIAL SELF-SIMILARITY METRICS

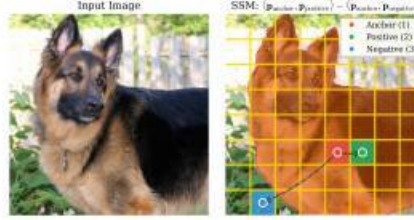


Figure 11: **Explaining semantic region self-similarity (SRSS) metric.** A visual exposition of the SRSS metric: for each image, we compute a mask by means of SAM2 and select anchor-positive-negative triplets. The SSM measures the difference in cosine similarity between anchor-positive pairs (within the mask) and anchor-negative pairs (outside the mask). Intuitively, larger values signify that patches within the same semantic region are more mutually similar than patches drawn from unrelated regions, and hence betoken a more faithful preservation of spatial structure.

Setup. Given an image $\mathcal{I}$ and a vision encoder $\mathcal{E}$, we extract patch tokens $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ with $T = H \times W$ and indices $t \in [T]$ arranged on an $H \times W$ lattice. Let $d : [T] \times [T] \rightarrow \mathbb{N}$ denote the Manhattan distance, and let the (cosine) self-similarity kernel be

$$K_{\mathcal{X}}(t, t') = \frac{\langle \mathbf{x}_t, \mathbf{x}_{t'} \rangle}{\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2} \in [-1, 1],$$

a standard local self-similarity measure (Shechtman & Irani, 2007).

- *Patch token representation:* $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ where $T = H \times W$ is the spatial grid of patches (patch index set $[T]$). Also let $d : [T] \times [T] \rightarrow \mathbb{N}$ be Manhattan distance on the $H \times W$ grid.
- *Self-similarity kernel:* $K : [T] \times [T] \rightarrow \mathbb{R}$ measuring self-similarity (Shechtman & Irani, 2007) between patch tokens. We use the cosine kernel $K_{\mathcal{X}}(t, t') = \langle \mathbf{x}_t, \mathbf{x}_{t'} \rangle / (\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2)$.
- *Spatial self-similarity metric:* a functional $m : \mathcal{K} \times \mathcal{D} \rightarrow \mathbb{R}$ that measures how the self-similarity K among patch tokens varies with the lattice distance d . Intuitively, larger values betoken stronger spatial organization, whereby near patches are more mutually similar than distant ones.

We now expound several elementary metrics for quantifying the spatial self-similarity structure of the target representations.

Local vs. Distant Similarity (LDS). We first consider a simple correlogram-contrast (local vs. distant) metric (Huang et al., 1997):

$$\text{LDS}(\mathcal{X}) = \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') < r_{\text{near}}] - \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') \geq r_{\text{far}}].$$

where r_{near} and r_{far} constitute the hyperparameters. By default we set $r_{\text{near}} = r_{\text{far}} = H/2$, having found the correlation to be robust to their precise values. Intuitively, larger values signify that, *on average*, proximate patches are more mutually similar than distant ones.

Correlation Decay Slope (CDS). Given the patch tokens $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$, we compute the spatial correlogram $g_{\mathcal{X}}(\delta) = \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') = \delta]$ for distances $\delta \in \{1, \dots, \Delta\}$. We then fit a least-squares line $\hat{g}_{\mathcal{X}}(\delta) \approx \alpha + \beta\delta$ and define:

$$\text{CDS}(\mathcal{X}) = -\hat{\beta}$$

where $\hat{\beta}$ denotes the fitted slope. Larger CDS values betoken a more rapid decay of similarity with distance, and hence a stronger spatial organization.

Semantic-Region Self-Similarity (SRSS). For each image, we first sample—by means of SAM2 (Ravi et al., 2024)—a binary mask $\mathbf{M} \in \mathbb{R}^{H \times W}$ containing a random object, downsampled to $H \times W$. We then select three categories of points: anchor, positive, and negative. The anchor and positive points are drawn from within the mask (and thus ought to share semantics), while the negative point is drawn from outside the mask (and is thus less related to the anchor). Conceptually, if the encoder’s feature representation preserves the spatial structure, the anchor and positive points should exhibit

higher cosine similarity, whereas the anchor and negative points should exhibit lower cosine similarity. Accordingly, we define the Spatial Structure Metric (SSM) as:

$$\text{SSM}(\mathbf{P}) = \mathbb{E}_{\text{anchor} \in \mathbf{M}} [\cos(\mathbf{p}_{\text{anchor}}, \mathbf{p}_{\text{pos}}) - \cos(\mathbf{p}_{\text{anchor}}, \mathbf{p}_{\text{neg}})]$$

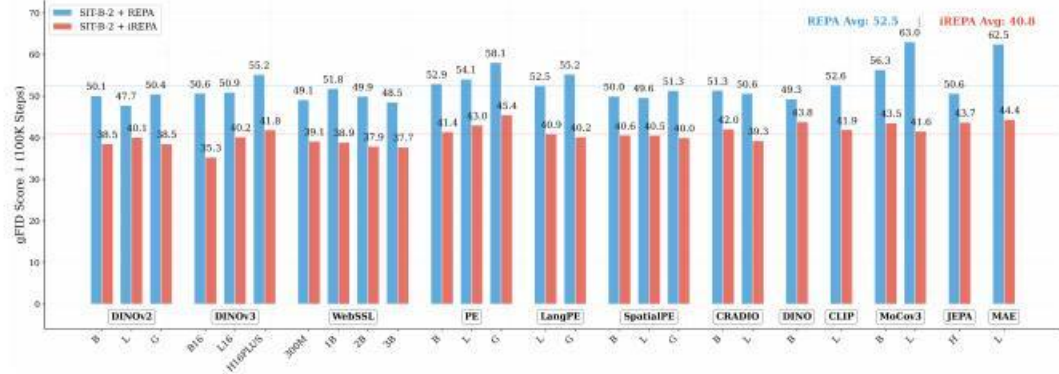
where positive points lie within Manhattan distance $d \leq r_{\text{near}}$ of the anchor and within the mask, and negative points lie at distance $d \geq r_{\text{far}}$ outside the mask.

RMS Spatial Contrast (RMSC). Finally, we consider a simple contrast metric that captures the spatial contrast among patch token representations. Given normalized features $\hat{\mathbf{x}}_t = \mathbf{x}_t / \|\mathbf{x}_t\|_2$ for each patch t , we compute:

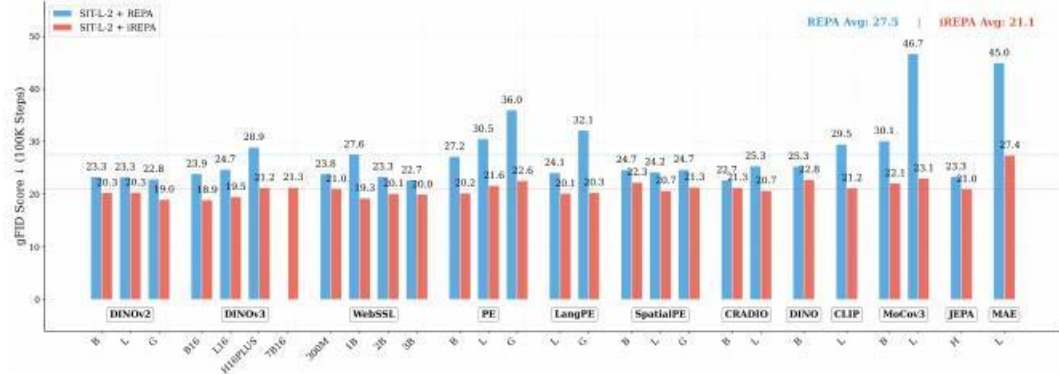
$$\text{RMSC}(\mathcal{X}) = \sqrt{\frac{1}{T} \sum_{t=1}^T \|\hat{\mathbf{x}}_t - \bar{\mathbf{x}}\|_2^2}$$

where $\bar{\mathbf{x}} = \frac{1}{T} \sum_{t=1}^T \hat{\mathbf{x}}_t$ denotes the mean of the normalized features across all spatial locations. Higher RMSC values connote greater spatial diversity in the feature representations, and hence preserved spatial structure, whereas lower values signal a more uniform feature distribution and thus a loss of spatial structure (which typically ensues when global information supplants spatial structure (Siméoni et al., 2025)).

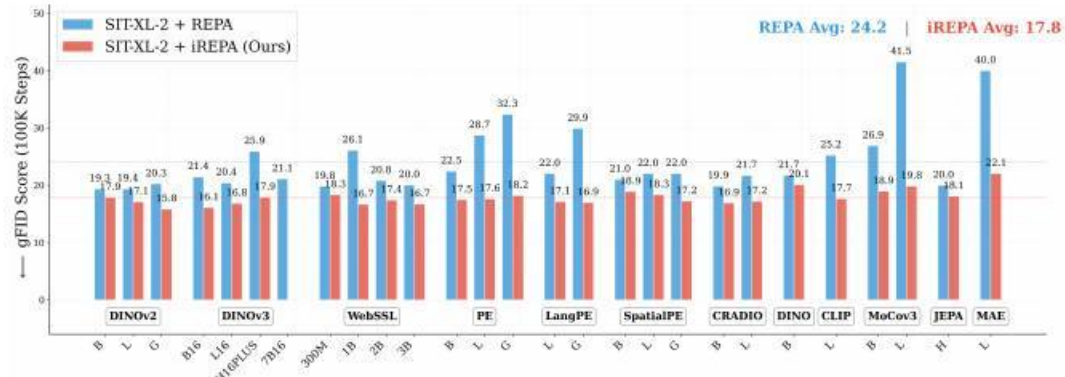
C VISION ENCODER VARIATION



(a) SiT-B-2 (130M parameters)



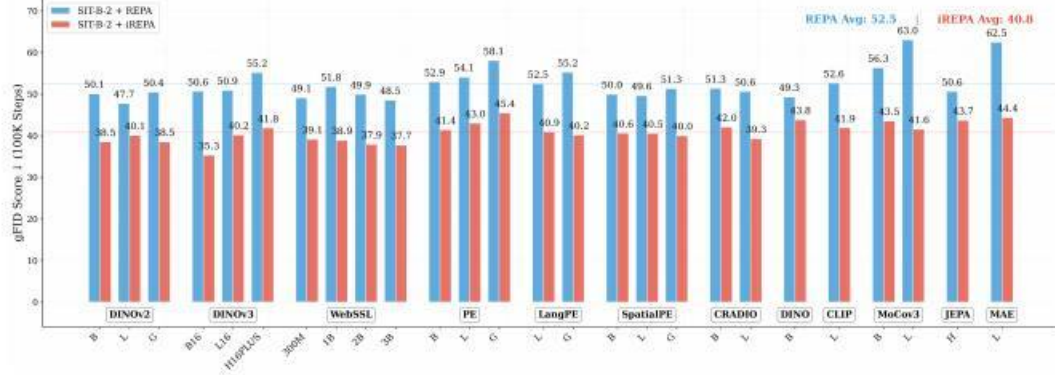
(b) SiT-L-2 (458M parameters)



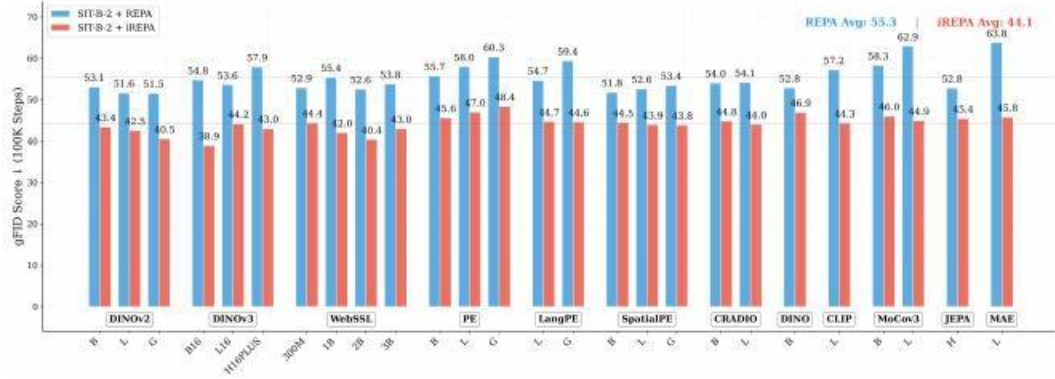
(c) SiT-XL-2 (675M parameters)

Figure 12: **Variation in vision encoders.** Ablation studies that exhibit the improvements afforded by iREPA across a range of vision encoders for the SiT-B-2, SiT-L-2, and SiT-XL-2 models. iREPA consistently enhances generation quality across all encoders and model sizes.

D ENCODER DEPTH VARIATION



(a) SiT-B-2 with 6 encoder layers



(b) SiT-B-2 with 8 encoder layers

Figure 13: **Effect of encoder depth on iREPA performance.** A comparison of SiT-B-2 performance at differing encoder depths (6 vs 8 layers) across a variety of vision encoders. The results attest to consistent improvements with iREPA irrespective of encoder depth, with marginally better performance obtained at 8 encoder layers. Both configurations exhibit significant FID improvements once spatial regularization is applied.

E COMPREHENSIVE RESULTS ACROSS VISION ENCODERS

E.1 SiT-B/2 RESULTS

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
CLIP-ViT-L	59.44	24.84	6.46	0.584	0.645
+iREPA	69.70	21.26	6.64	0.601	0.644
DINOv2-B	64.92	22.75	6.54	0.593	0.653
+iREPA	70.88	21.40	6.77	0.597	0.653
DINOv2-L	66.65	22.46	6.53	0.594	0.650
+iREPA	72.41	20.99	6.84	0.600	0.650
DINOv3-B16	62.57	23.91	6.77	0.584	0.647
+iREPA	75.48	19.65	6.68	0.606	0.649
DINOv3-H16+	61.91	24.82	6.57	0.579	0.645
+iREPA	71.76	21.80	6.84	0.592	0.655
DINO-B	58.37	24.71	6.32	0.586	0.635
+iREPA	57.55	24.41	6.43	0.587	0.638
LangPE-G	58.42	25.37	6.32	0.583	0.641
+iREPA	66.88	21.62	6.65	0.604	0.649
MoCov3-B	57.61	25.60	6.25	0.579	0.640
+iREPA	61.41	22.80	6.46	0.596	0.639
PE-B	58.86	25.78	6.44	0.58	0.64
+iREPA	69.50	21.15	6.66	0.601	0.659
PE-G	56.09	26.84	6.45	0.568	0.641
+iREPA	70.43	21.52	6.58	0.598	0.651
SpatialPE-B	59.19	24.28	6.37	0.586	0.646
+iREPA	61.50	22.71	6.41	0.604	0.644
SpatialPE-L	62.13	23.87	6.51	0.59	0.65
+iREPA	68.83	21.07	6.59	0.607	0.651
WebSSL-1B	63.20	23.48	6.33	0.593	0.646
+iREPA	73.54	19.74	6.63	0.612	0.646
WebSSL-2B	66.56	22.27	6.69	0.596	0.654
+iREPA	72.13	20.47	6.89	0.609	0.648
WebSSL-300M	62.06	23.64	6.48	0.590	0.646
+iREPA	66.83	21.46	6.71	0.606	0.653
WebSSL-3B	65.91	22.55	6.53	0.598	0.654
+iREPA	71.98	20.64	6.71	0.607	0.650

Table 5: **SiT-B/2 performance across vision encoders at 400K iterations.** iREPA consistently improves generation quality. All baselines are reported using vanilla REPA (Yu et al., 2024) for training.

E.2 SiT-L/2 RESULTS

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
CLIP-ViT-L	107.7	10.6	5.20	0.682	0.643
+iREPA	115.8	9.41	5.15	0.689	0.653
DINOv2-B	113.3	9.61	5.11	0.684	0.653
+iREPA	113.9	9.36	5.11	0.690	0.655
DINOv3-B16	115.3	9.41	5.24	0.687	0.651
+iREPA	118.1	9.06	5.25	0.685	0.650
DINOv3-H16+	110.7	10.6	5.32	0.677	0.652
+iREPA	120.5	9.64	5.29	0.681	0.663
LangPE-G	102.8	11.2	5.13	0.684	0.637
+iREPA	112.7	9.37	5.02	0.690	0.653
MoCov3-B	102.1	11.0	5.14	0.685	0.635
+iREPA	104.7	10.1	5.00	0.690	0.641
PE-G	98.38	12.6	5.31	0.670	0.646
+iREPA	117.1	9.65	5.33	0.683	0.651
WebSSL-1B	107.9	10.5	5.13	0.686	0.644
+iREPA	121.5	8.62	5.01	0.696	0.653

Table 6: **SiT-L/2 performance across vision encoders at 400K iterations.** iREPA consistently improves generation quality across all encoders, with especially pronounced gains in the FID and IS metrics. All baselines are reported using vanilla REPA (Yu et al., 2024) for training.

F ADDITIONAL EXPERIMENTAL RESULTS

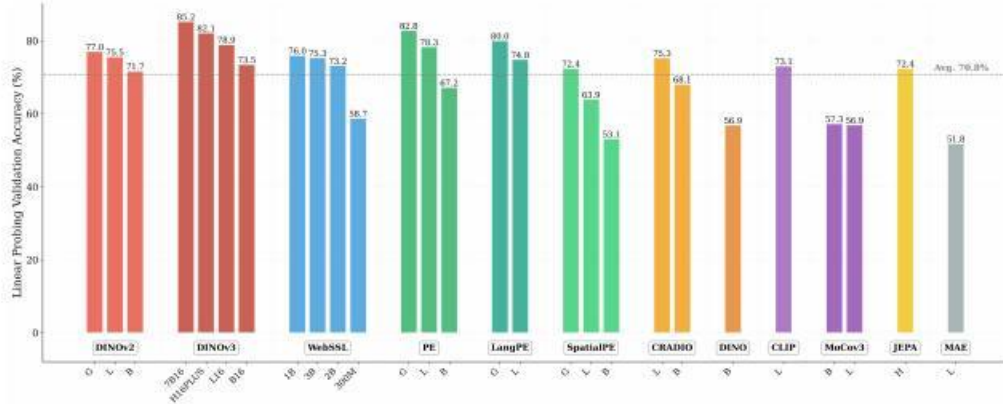


Figure 14: **Global information in mean patch tokens.** We find that, beyond the [CLS] token, the mean of the patch tokens likewise encodes substantial global semantic information. While this augments global performance, it simultaneously diminishes the contrast among individual patch tokens, thereby potentially impeding the transfer of spatial structure. We find that a portion of this global information (embodied in the mean of the patch tokens) may be excised so as to improve spatial structure transfer during representation alignment (§4).

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
DINOv2-B	262.97	1.98	4.60	0.799	0.610
+iREPA (Ours)	268.79	1.93	4.59	0.799	0.600
DINOv3-B	261.18	1.99	4.58	0.799	0.609
+iREPA (Ours)	272.41	1.89	4.58	0.799	0.600
WebSSL-1B	250.53	2.24	4.61	0.809	0.580
+iREPA (Ours)	271.59	1.90	4.58	0.798	0.609
PE-G	238.37	2.44	4.57	0.805	0.585
+iREPA (Ours)	275.36	1.93	4.59	0.796	0.606

Table 7: **Generation quality with classifier-free guidance.** A comparison of REPA against iREPA under CFG (scale 2.0) across a range of vision encoders. All experiments employ SiT-XL/2 trained for 400K iterations with 250 sampling steps. iREPA consistently improves both the IS and FID metrics across every encoder.

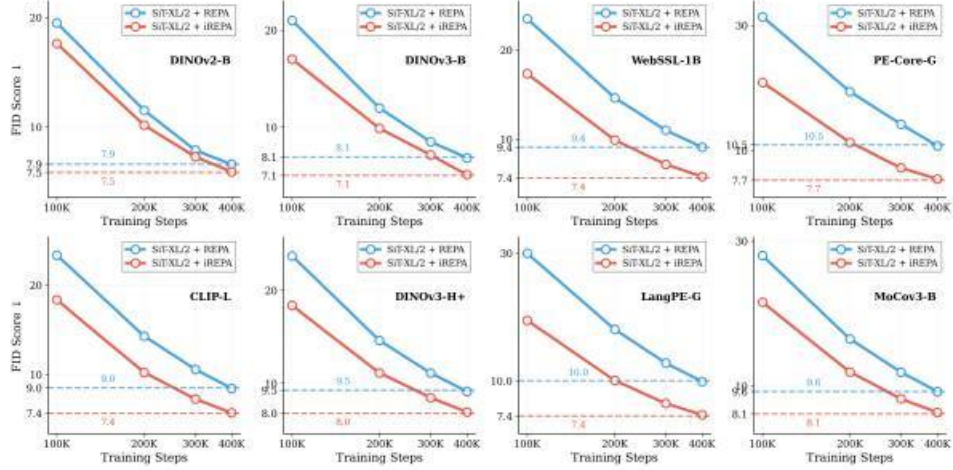


Figure 15: Accentuating spatial features helps consistently improve convergence speed. Results for SiT-XL/2 with REPA and iREPA.

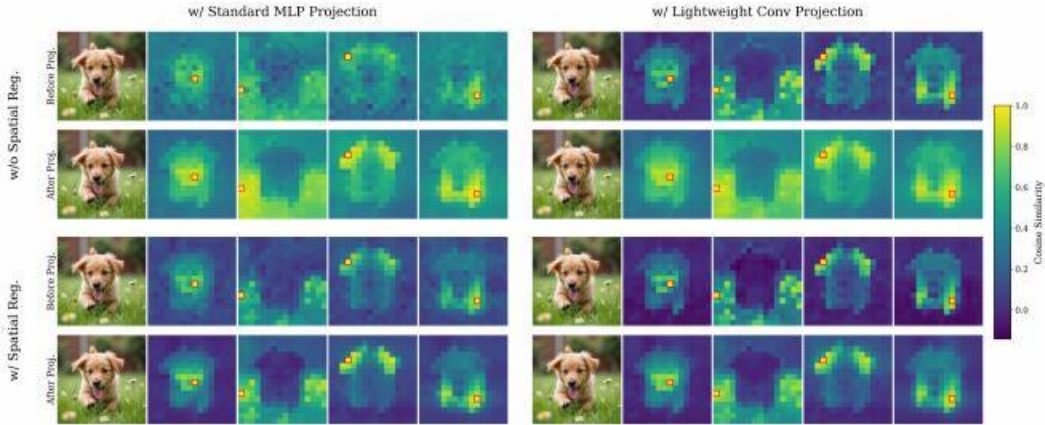


Figure 16: Visualizing the impact of two elementary improvements that enhance spatial feature transfer. First, we find that the standard MLP projection layer (top-left) forfeits spatial information in the course of transferring features from the pretrained encoder (after projection) to the diffusion model (before projection). Substituting a simpler convolutional layer, by contrast, more faithfully preserves the transfer of spatial information (top-right). Second, we observe that vision encoder features frequently possess limited spatial contrast (§2), whereby the tokens of one semantic region (e.g., a dog) come to exhibit appreciable cosine similarity with unrelated tokens (e.g., the background). We remedy this with a simple spatial regularization layer that accentuates the spatial contrast of the learned representation, thereby securing better generation performance. The best results are obtained when both are employed in conjunction (see Table 2).

G IMPLEMENTATION DETAILS

Training setup. We adhere to the same setup as REPA (Yu et al., 2024) unless otherwise specified. All training is conducted on the ImageNet (Deng et al., 2009) training split. For preprocessing, we adopt the protocol of ADM (Dhariwal & Nichol, 2021), whereby each image is center-cropped and resized to a resolution of 256×256 . We employ the `stabilityai/sd-vae-ft-mse` VAE (AI, n.d.) throughout diffusion training and inference. For the spatial normalization layer we use $\gamma \in [0.6, 0.8]$, and for the projection layer we use a convolutional layer with kernel size 3 and padding 1. For optimization, we employ AdamW (Kingma & Ba, 2014; Loshchilov, 2017) with a constant learning rate of 1×10^{-4} and a global batch size of 256. During training, we adopt `bfloat16` mixed precision together with `torch.compile` to expedite training, and we apply gradient clipping and an exponential moving average (EMA) of the generative model to stabilize optimization.

For the REPA-E (Leng et al., 2025a) and JiT (Li & He, 2025a) experiments, we use the official open-source implementations. For the MeanFlow (Geng et al., 2025) experiments, we obtained the implementation (REPA included) from the original authors and adapted it to incorporate the two elementary modifications constituting iREPA (§4).

Evaluation. For the evaluation of image generation, we adhere strictly to the ADM setup (Dhariwal & Nichol, 2021). We report generation quality in terms of the Fréchet inception distance (gFID) (Heusel et al., 2017), structural FID (sFID) (Nash et al., 2021), inception score (IS) (Salimans et al., 2016), and precision (Prec.) and recall (Rec.) (Kynkäänniemi et al., 2019), all measured over 50K generated images. For sampling, we follow the approach of REPA (Yu et al., 2024), employing the SDE Euler-Maruyama sampler with 250 steps. For JiT (Li & He, 2025a), we use 50 inference steps, in keeping with the official implementation.

H MORE DISCUSSION ON RELATED WORK

Classical spatial features in computer vision. Spatial feature extraction has long been foundational to computer vision. Classical methods such as SIFT (Lowe, 1999), HOG (Dalal & Triggs, 2005), SURF (Bay et al., 2006), and ORB (Rublee et al., 2011) furnish robust local descriptors for tasks ranging from object detection to image matching. Whereas these handcrafted features excel at capturing local spatial patterns and geometric invariances, recent work in generative modeling has fixated predominantly upon the representations of modern self-supervised methods that demonstrate strong global classification performance, such as DINOv2 (Oquab et al., 2024) and CLIP (Radford et al., 2021). Our findings intimate a different perspective: since the preservation of spatial structure is critical to generation quality, even classical spatial features could conceivably improve diffusion training once suitably aligned. This underscores the promise of harnessing the full spectrum of spatial feature extractors—from traditional handcrafted descriptors to modern learned representations—provided that they sustain strong spatial coherence.

Pretrained visual encoders for generative models. Pretrained visual encoders have buttressed generative models in several capacities: as discriminators to accelerate GAN convergence (Goodfellow et al., 2020; Sauer et al., 2021; 2022; 2023; Radford et al., 2021), as teachers in the adversarial distillation of diffusion models (Sauer et al., 2024; Kang et al., 2024), and, more recently, as alignment targets. In GANs (Goodfellow et al., 2020), pretrained features have not only improved convergence speed but also enabled scaling to large datasets, as evidenced by StyleGAN-XL (Sauer et al., 2022) and StyleGAN-T (Sauer et al., 2023) employing CLIP (Radford et al., 2021) features. For diffusion models, adversarial distillation exploits pretrained encoders to steer student networks toward higher-fidelity samples, with manifest improvements in perceptual quality. REPA (Yu et al., 2024), in particular, aligns diffusion features with external encoders, demonstrating that representation alignment can improve both generation convergence and quality. Building upon this line of work, we concentrate not merely on alignment but specifically on the preservation of spatial structure rather than on discriminative capability.

Denosing transformers. Transformer architectures have become the predominant backbone for scalable generative modeling, encompassing a variety of formulations including diffusion transformers (Peebles & Xie, 2023) and flow-matching variants (Ma et al., 2024). Recent architectures such as GenTron (Chen et al., 2023) scale transformers beyond 3B parameters for text-to-image synthesis, while U-ViT (Bao et al., 2023) demonstrates that pure transformer backbones, bereft of U-Net

inductive biases, can nonetheless attain competitive performance. ARDiT (Tang et al., 2024) and DART (Jing et al., 2024) explore autoregressive formulations that marry denoising with sequential generation, thereby affording flexible trade-offs between quality and speed. These architectural advances notwithstanding, training such models from scratch remains computationally onerous, frequently demanding millions of iterations to reach satisfactory generation quality. Although representation alignment methods such as REPA (Yu et al., 2024) have shown that pretrained features can dramatically accelerate convergence, the mechanisms underlying this improvement have hitherto remained obscure. Our analysis reveals that the principal benefit derives from the preservation of spatial structure rather than from semantic alignment, which explains why certain encoders furnish stronger acceleration than others and, in turn, guides the design of more effective alignment strategies. While (Wang et al., 2025c) propose improvements to REPA training through early stopping, they hypothesize that “REPA predominantly distills global semantic information while leaving structural information untapped.” In contrast, our analysis reveals that spatial structure (not global semantic information) already plays a very significant role in the effectiveness of REPA (§2, §3). Thus, while we indeed find that spatial structure remains underexploited (§4) in REPA, we surprisingly find that majority of improvements in REPA are already coming from introducing an inductive bias for spatial structure (not global information).

Denoising as self-supervised learning task. The nexus between denoising and representation learning has been examined from a multiplicity of vantage points. Early work by Abstreiter et al. (2021) extended diffusion objectives to representation learning, demonstrating that denoising naturally acquires meaningful features. SODA (Hudson et al., 2024) introduces a diffusion model equipped with an information bottleneck so as to learn compact representations. Chen et al. (2024) deconstruct diffusion models and find that a simple denoising autoencoder suffices for strong self-supervised performance, while Wang & He (2025) introduce a dispersive loss to promote internal representation diversity in diffusion models, thereby improving generative performance without recourse to external encoders. In a kindred vein, Jiang et al. (2025) propose Self-Representation Alignment (SRA), which aligns a diffusion transformer’s latent features across noise levels so as to provide self-guidance without an auxiliary model. Collectively, these works establish denoising as a fundamental self-supervised task that naturally encourages the learning of robust features. Our findings complement this view by demonstrating that, when diffusion models are aligned with strong spatial representations drawn from self-supervised encoders, both their generative and discriminative capabilities improve—an observation that implicates the preservation of spatial structure as a key factor in this synergy.

Scaling self-supervised vision encoders. Recent years have witnessed remarkable progress in scaling self-supervised vision models to unprecedented sizes and datasets. DINOv3 (Siméoni et al., 2025) trains a 7B-parameter ViT on 1.7 billion unlabeled images by aligning the representations of distinct augmentations. WebSSL (Fan et al., 2025) demonstrates that visual models trained on upwards of 2 billion images can rival language-supervised models such as CLIP (Radford et al., 2021) on vision-language tasks without any language supervision. C-RADIO (Heinrich et al., 2024) amalgamates multiple teacher models through distillation to yield versatile encoders that excel across diverse visual domains. I-JEPA (Assran et al., 2023) explores predictive architectures that learn by predicting masked regions in representation space rather than by reconstructing pixels directly, while SAM (Ravi et al., 2024) specializes in promptable segmentation through large-scale supervised training. The Perception Encoder family (Bolya et al., 2025; Cho et al., 2025) shows that intermediate features frequently outperform final representations on dense prediction tasks. Although these models are typically evaluated on global tasks such as image classification, our work brings to light a crucial insight: strong performance on discriminative benchmarks does not necessarily translate into better generation quality. On the contrary, we find that it is encoders preserving spatial structure—irrespective of their classification accuracy—that confer the greatest benefit upon diffusion training. This suggests that the evaluation metrics for self-supervised encoders warrant reconsideration when generative applications are the objective, with spatial coherence being no less important than semantic understanding.

I NOTE ON LLM USAGE

We employ GPT-5 (OpenAI, 2025) to deliberate over various formulations of the spatial structure metrics discussed in §3. Furthermore, all figures in the paper are generated directly from our experiment logs and checkpoints by means of Claude-Code (Anthropic, 2025). We additionally enlist

1404 LLM assistance for the retrieval and articulation of related work in Appendix H, and we make some
1405 use of Cursor to aid in the writing of the paper.
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J ADDITIONAL DISCUSSION ON SPATIAL STRUCTURE METRICS

Spatial structure metrics can be used to measure the representation gap. Yu et al. (2024) employ linear probing to gauge the representation gap, invoking the attendant increase in validation accuracy to account for the efficacy of REPA. We find that spatial structure metrics (SSM) may likewise be used to quantify the representation gap and to explain the improvements observed across models. As Figure 17 shows, representation alignment serves to close the gap between the SSM performance of a pretrained encoder such as DINOv2 and that of the diffusion features.

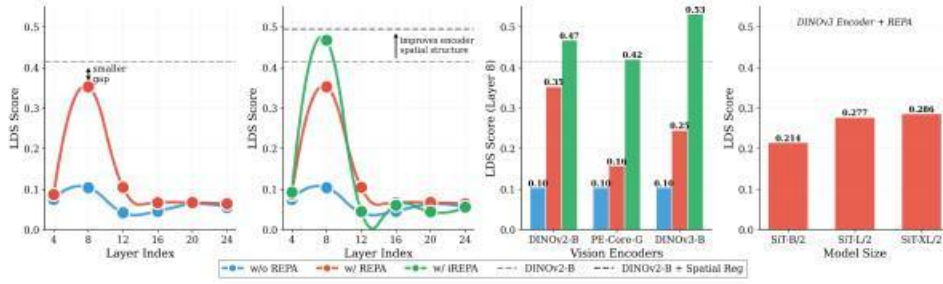


Figure 17: Representation alignment as bridging the spatial feature gap. a) We find that representation alignment may also be construed as bridging the spatial feature gap between the patch features of the diffusion model and those of the vision encoder. b) iREPA (§4) accentuates the spatial features of the vision encoder—at the cost of some global information—thereby securing better generation performance. c) iREPA consistently improves performance across a range of vision encoders. d) The spatial structure improvements scale with model size. All results are reported with SiT-XL/2 at 400k iterations.

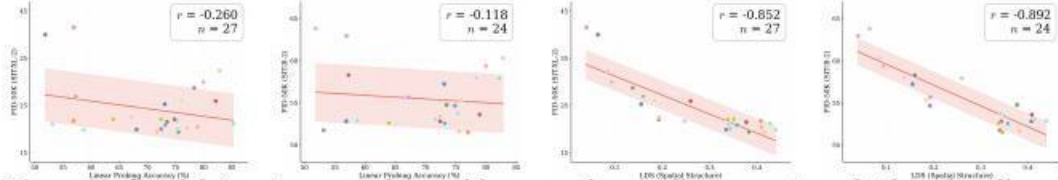


Figure 18: Spatial structure correlates with generation quality more strongly than does linear probing. A correlation analysis across 27 vision encoders. **Left two:** The correlation of linear probing accuracy with FID, for SiT-XL-2 (Pearson $r = -0.26$) and SiT-B-2 ($r = -0.12$), is weak. **Right two:** The correlation of spatial structure (LDS) with FID, for SiT-XL-2 ($r = -0.85$) and SiT-B-2 ($r = -0.89$), is strong. See Figure 10 in the Appendix for detailed plots annotated with encoder labels.

K ADDITIONAL ANECDOTAL COMPARISONS

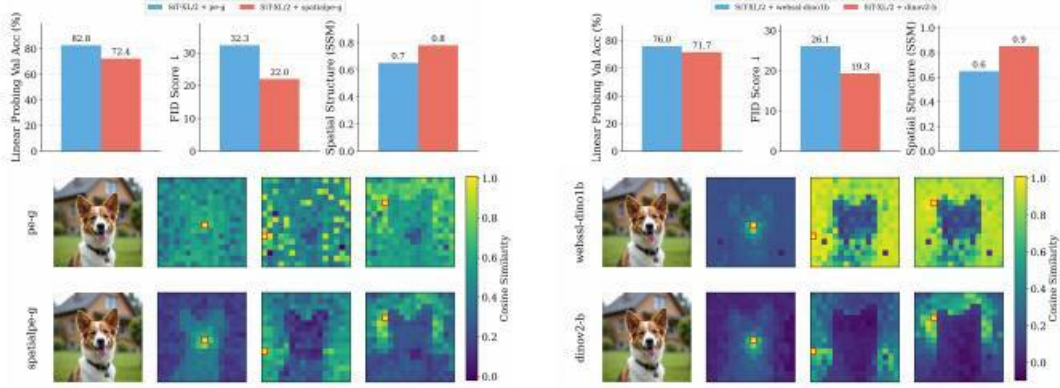


Figure 19: **Motivating anecdotes from recent SSL encoders — spatial structure matters.** **Top row:** A comparison of metrics that lays bare the inverse relationship between ImageNet accuracy and generation quality. **Left:** PE-g attains higher accuracy (82.8% vs 72.4%) yet a worse FID (32.3 vs 22.0) than Spatial-PE-g, owing to its much weaker spatial structure (LDS: 0.1 vs 0.4). **Right:** WebSSL-dino1b exhibits higher accuracy (76.0% vs 71.7%) yet a worse FID (26.1 vs 19.3) than DINOv2-b, on account of its weaker spatial structure (LDS: 0.2 vs 0.4). **Bottom row:** Spatial cosine similarity visualizations confirm that the encoders yielding better generation (Spatial-PE-g, DINOv2-b) sustain clear spatial coherence, whereas those optimized for classification (PE-g, WebSSL) forfeit spatial structure.

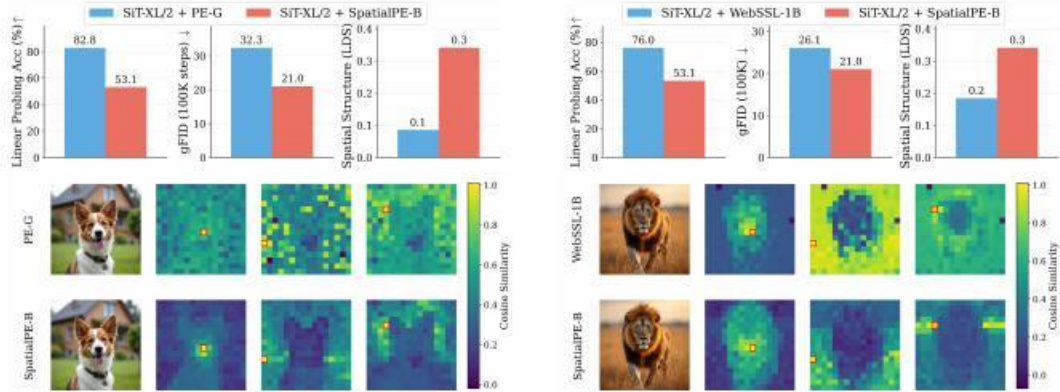


Figure 20: **Motivating examples — spatial structure matters.** **Top:** A comparison of metrics that lays bare the inverse relationship between ImageNet accuracy and generation quality. **Left:** PE_{core-G}, its significantly higher validation accuracy (82.8% vs. 53.1%) notwithstanding, exhibits worse generation quality than PE_{Spatial-B} (Bolya et al., 2025). **Right:** In like manner, WebSSL-1B (Fan et al., 2025) attains much stronger global performance (76.0% vs. 53.1%) yet weaker generation. **Bottom:** We find that spatial structure instead furnishes a more reliable predictor of generation quality than does global performance. See §3 for the spatial structure metric. All results are reported at 100K using SiT-XL/2 and REPA.

L FURTHER DISCUSSION

Role of Spatial Normalization We visualize the effect of spatial normalization upon vision encoder representations. Vision encoders can possess substantial global components, or overlays, that curtail the contrast among spatial tokens. The tokens of one semantic region, for instance, can be highly correlated with tokens in unrelated regions (e.g., the background), thereby eroding the spatial distinctiveness of the features. Spatial normalization excises this global overlay so as to heighten the contrast among different spatial tokens, enabling the model to preserve local spatial structure more faithfully while attenuating the dominance of the global information that can interfere with generation quality.

The following figures illustrate this effect across a range of examples, exhibiting how spatial normalization transforms the feature representations so as to accentuate spatial structure:

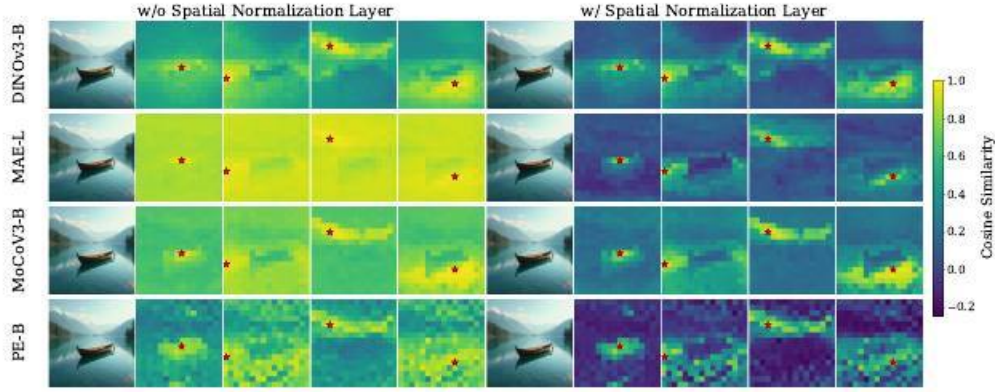


Figure 21: Visualizing the impact of spatial normalization (Example 1). The heatmaps depict token similarity patterns before and after spatial normalization. In the absence of normalization (left), the global components engender high correlations across unrelated regions; with spatial normalization (right), local spatial structure is accentuated and global interference attenuated, yielding more distinct semantic boundaries.

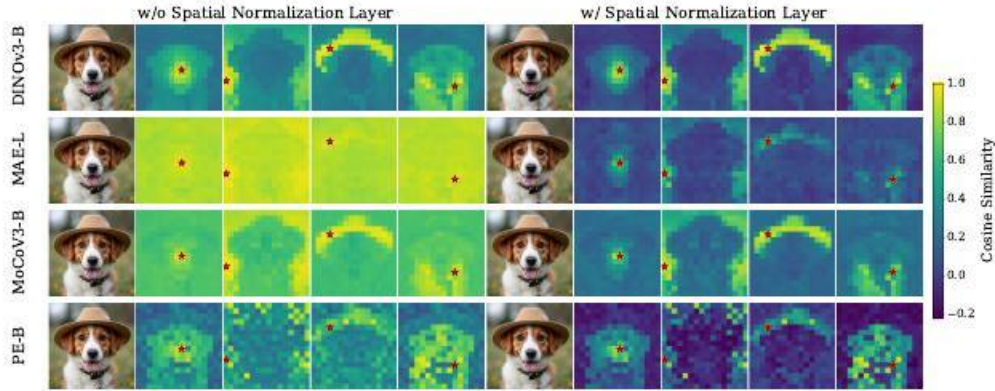


Figure 22: Visualizing the impact of spatial normalization (Example 3). The heatmaps depict token similarity patterns before and after spatial normalization. In the absence of normalization (left), the global components engender high correlations across unrelated regions; with spatial normalization (right), local spatial structure is accentuated and global interference attenuated, yielding more distinct semantic boundaries.

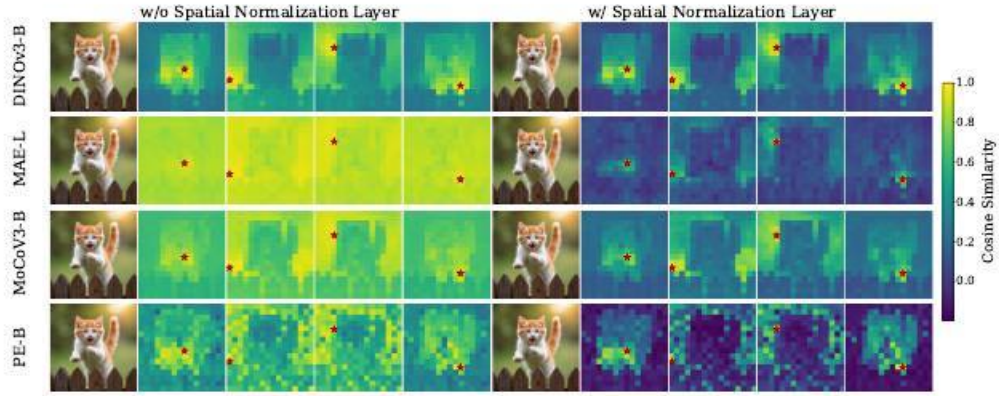


Figure 23: Visualizing the impact of spatial normalization (Example 2). Feature similarity maps demonstrate how spatial normalization heightens spatial contrast. The original features (left) exhibit a global overlay that blurs the distinction between foreground and background regions; after normalization (right), the spatial tokens grow more locally coherent, with a clearer separation between the different semantic regions.

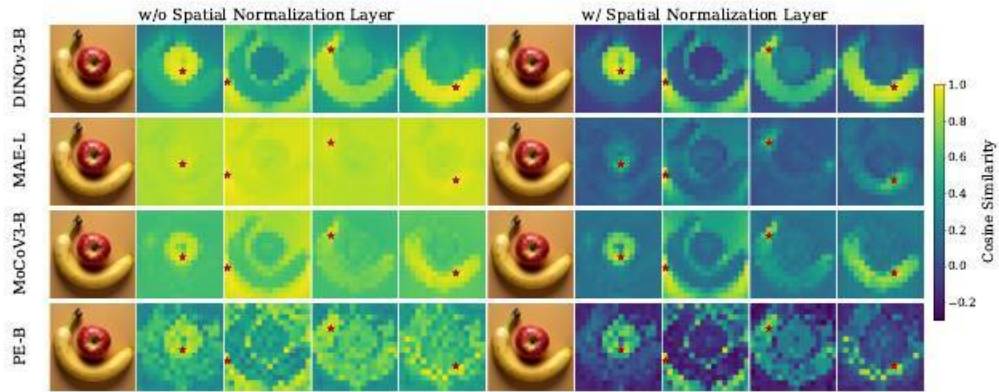


Figure 24: **Visualizing the impact of spatial normalization** (Example 4). The heatmaps depict token similarity patterns before and after spatial normalization. In the absence of normalization (left), the global components engender high correlations across unrelated regions; with spatial normalization (right), local spatial structure is accentuated and global interference attenuated, yielding more distinct semantic boundaries.

M ADDITIONAL RESULTS

M.1 ADDITIONAL RESULTS AT HIGHER RESOLUTIONS

Setup. We conduct further experiments on ImageNet-512 (Deng et al., 2009) in order to evaluate the generalization of the proposed spatial improvements at higher resolutions. We adhere to the same setup as REPA (Yu et al., 2024) and report results across a range of pretrained encoders (DINOv2, DINOv3, WebSSL, PE, etc.). All results are reported using SiT-XL and SiT-B with 50 NFE at inference and without classifier-free guidance. Fig. 25 presents the results with REPA both before and after the application of the spatial improvements (iREPA). We observe that the spatial improvements generalize to higher resolutions as well, and that consistent gains obtain across the several pretrained encoders (DINOv2, DINOv3, WebSSL, PE, etc.).

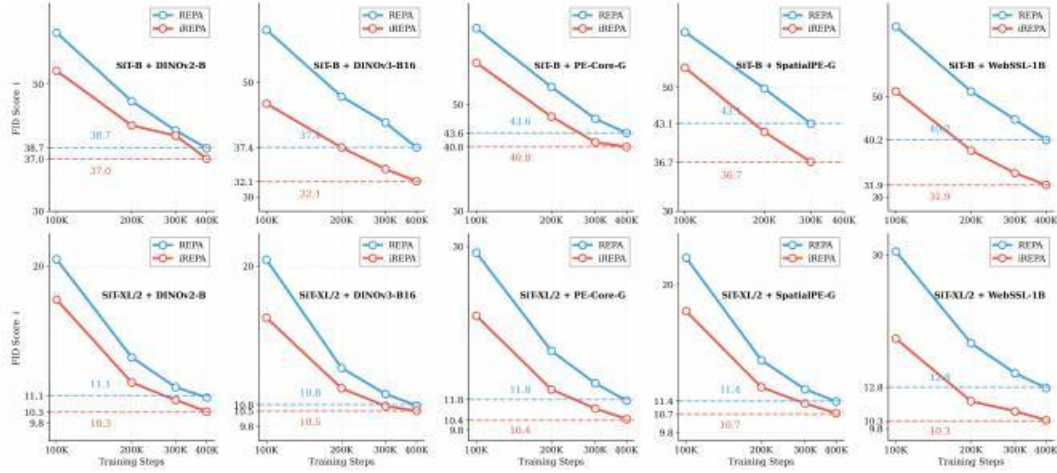


Figure 25: **Convergence results at 512x512 resolution.** Accentuating spatial features consistently accelerates convergence across resolutions, for both ImageNet 256 (Figure 7) and ImageNet 512 (above).

M.2 ADDITIONAL RESULTS ON MULTIMODAL T2I TASKS

Setup. To further probe the generalizability of the proposed spatial improvements beyond ImageNet, we additionally perform extensive experiments on multimodal T2I tasks. Following REPA (Yu et al., 2024), we adopt MMDiT (Esser et al., 2024) as the diffusion backbone and apply REPA with a variety of pretrained vision encoders (DINOv2, CLIP, WebSSL, PE, etc.). As in REPA (Yu et al., 2024), all models are trained for 150K steps with a batch size of 256 and evaluated using an ODE sampler with 50 NFE. Fig. 26 presents the results with REPA both before and after the application of the spatial improvements (iREPA). We observe that the spatial improvements generalize to multimodal T2I tasks as well, and that consistent gains obtain across the several pretrained encoders (DINOv2, CLIP, WebSSL, PE, etc.).

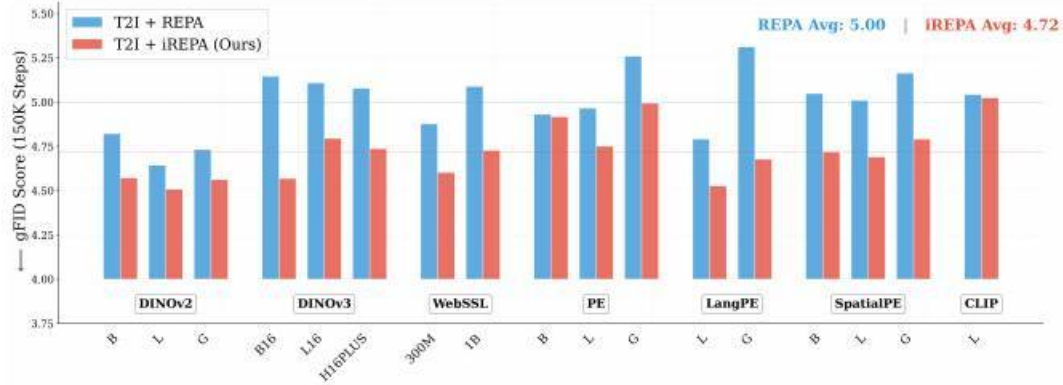


Figure 26: **Text-to-image generation across encoder variants.** Accentuating the transfer of spatial features from the target representation to the diffusion features consistently accelerates convergence for both ImageNet (see §4) and multimodal T2I tasks (above). Consistent gains, moreover, obtain across the several pretrained encoders (DINOv2, CLIP, WebSSL, PE, etc.).

M.3 CORRELATION ANALYSIS WITHOUT OUTLIERS (MoCOv3-L AND MAE-L)

We repeat the correlation analysis of Section 3 after excising MoCOv3-L and MAE-L, which we identified as outliers. Figure 28 presents the updated correlations across model sizes. We observe that spatial structure continues to correlate far more strongly with generation performance than does linear probing accuracy. Intriguingly, once the outliers are removed, linear probing in fact exhibits a small positive correlation with gFID (that is, as linear probing performance rises, generation deteriorates). This trend accords with the observations of Sec. 2, wherein target representations of higher global semantic performance (linear probing accuracy) frequently deliver comparable or worse generation performance under representation alignment (REPA).

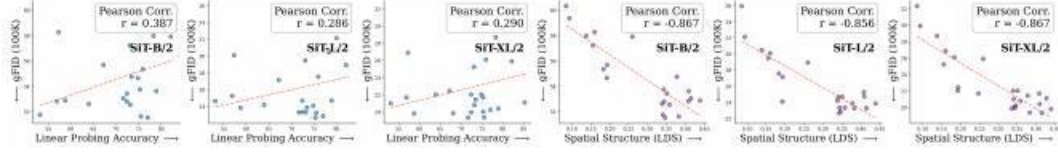


Figure 27: Correlation analysis without outliers. Across all model sizes (B, L, XL), spatial structure continues to correlate far more strongly (Pearson $|r| > 0.85$) with generation performance than does linear probing accuracy (Pearson $|r| < 0.38$) once the outliers (MoCOv3-L and MAE-L) are removed. Intriguingly, after the removal of the outliers, linear probing in fact exhibits a small positive correlation with gFID (that is, as linear probing performance rises, generation deteriorates). This trend accords with the observations of Sec. 2, wherein target representations of higher global semantic performance (linear probing accuracy) frequently deliver comparable or worse generation performance under representation alignment (REPA).

M.4 ADDITIONAL QUALITATIVE RESULTS

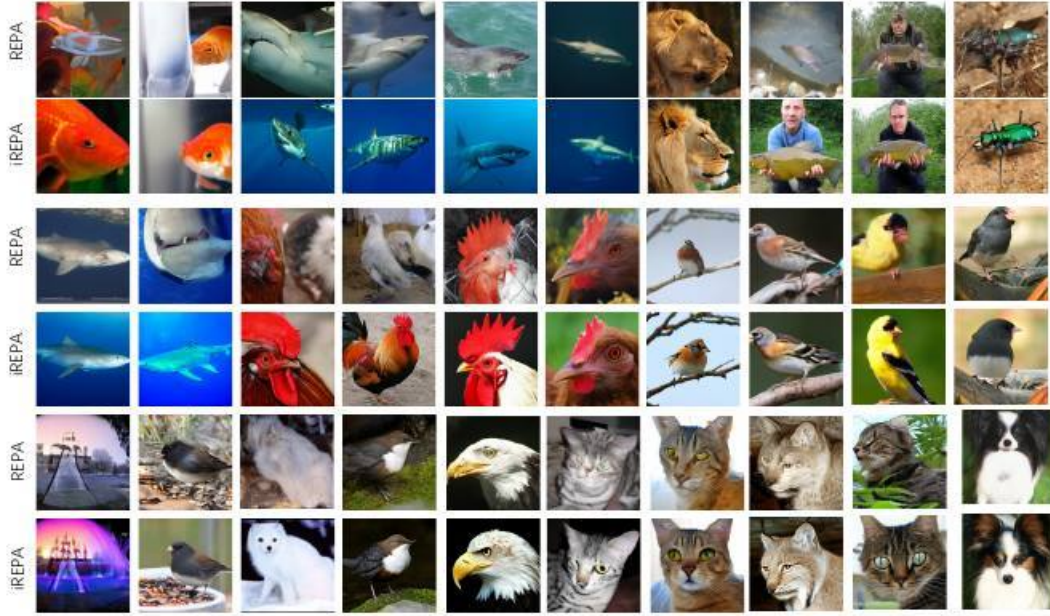


Figure 28: **Additional Qualitative Results** juxtaposing generation outputs before and after the application of the spatial improvements with REPA. All results are reported using PE-G (Bolya et al., 2025) as the pretrained vision encoder, at 400K steps (80 epochs) and with a classifier-free guidance scale of 4.0. In consonance with the quantitative improvements (§4), we observe that the spatial improvements (iREPA) likewise enhance the visual quality and coherence of the generated outputs.

M.5 ADDITIONAL RESULTS WITH ALTERNATIVE EVALUATION METRICS

Beyond the traditional evaluation metrics (Inception Score, FID, sFID, Precision, Recall), we further verify the robustness of the proposed findings under alternative measures of generation quality (Stein et al., 2023; Yang et al., 2023; Kynkäänniemi et al., 2023; Jayasumana et al., 2024). In particular, we employ the CMMD metric (Jayasumana et al., 2024) together with the PyTorch implementation of (Paul, 2024), taking the standard reference set from the OpenAI ADM evaluation suite (Dhariwal & Nichol, 2021) as reference images. We next corroborate the robustness of the key findings from both §3 and §4 under the CMMD metric.

Spatial structure correlates much higher with generation performance than linear probing.

To interrogate the robustness of the analysis in §3, we repeat the large-scale correlation analysis across vision encoders using the CMMD metric (Jayasumana et al., 2024). The results are reported in Figure 29; all results employ SiT-B/2 (100K steps) and REPA. Every spatial metric correlates far more strongly (Pearson $|r| > 0.88$) with generation performance (CMMD) than does linear probing (Pearson $|r| = 0.074$), thereby demonstrating that the key empirical findings of §3 are robust to the choice of evaluation metric (whether FID or CMMD).

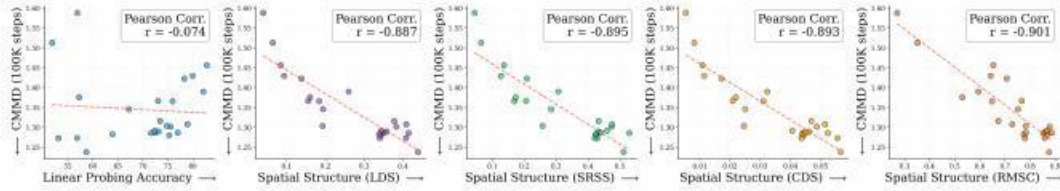


Figure 29: **Correlation analysis with CMMD metric.** We repeat the correlation analysis of §3 using the CMMD metric (Jayasumana et al., 2024) in place of gFID. All results employ SiT-B/2 (100K steps) and REPA. Every spatial metric correlates far more strongly (Pearson $|r| > 0.88$) with generation performance (CMMD) than does linear probing (Pearson $|r| = 0.074$). This demonstrates that the key empirical findings of §3 are robust to the choice of evaluation metric.

Accentuating transfer of spatial features helps improve generation quality. We next examine the generalizability of the analysis in §4 under the CMMD (Jayasumana et al., 2024) metric, in addition to the traditional evaluation metrics (gFID, sFID, IS, etc.). The results are reported in Table 8. Across the range of vision encoders, accentuating the transfer of spatial features (iREPA) serves to accelerate convergence under both CMMD and the traditional evaluation metrics (IS, FID, sFID, Prec., Recall). Consistent improvements, moreover, are observed both with and without classifier-free guidance (CFG).

Vision Encoder	Steps	w/o CFG						w/ CFG					
		CMMD↓	IS↑	FID↓	sFID↓	Prec.↑	Rec.↑	CMMD↓	IS↑	FID↓	sFID↓	Prec.↑	Rec.↑
DINOv2-B	100K	0.702	69.20	19.3	5.89	0.64	0.61	0.529	157.2	6.35	5.91	0.77	0.54
+iREPA	100K	0.652	77.92	16.9	6.26	0.66	0.61	0.484	179.3	5.15	6.23	0.78	0.54
DINOv2-B	400K	0.455	127.4	7.76	5.06	0.70	0.66	0.320	263.0	1.98	4.60	0.80	0.61
+iREPA	400K	0.438	128.6	7.52	4.89	0.71	0.65	0.310	268.8	1.93	4.59	0.80	0.60
DINOv3-B	100K	0.749	63.64	21.4	6.14	0.63	0.60	0.571	144.0	7.57	6.09	0.76	0.53
+iREPA	100K	0.651	78.79	16.2	6.14	0.66	0.61	0.481	181.9	4.87	6.10	0.78	0.55
DINOv3-B	400K	0.474	126.7	8.10	5.06	0.70	0.66	0.336	261.2	1.99	4.58	0.80	0.61
+iREPA	400K	0.441	132.9	7.13	4.93	0.71	0.66	0.314	272.4	1.89	4.58	0.80	0.60
WebSSL-1B	100K	0.825	53.87	25.5	6.57	0.61	0.59	0.622	124.0	9.59	6.37	0.76	0.52
+iREPA	100K	0.653	77.47	16.6	6.18	0.66	0.61	0.484	177.6	5.09	6.11	0.79	0.54
WebSSL-1B	400K	0.512	116.9	9.39	5.14	0.70	0.64	0.354	250.5	2.24	4.61	0.81	0.58
+iREPA	400K	0.445	130.8	7.48	4.91	0.70	0.65	0.311	271.6	1.90	4.58	0.80	0.61
PE-Core-G	100K	0.922	42.74	32.3	6.70	0.57	0.59	0.714	97.2	14.1	6.56	0.71	0.53
+iREPA	100K	0.697	75.01	18.1	6.03	0.64	0.61	0.525	176.8	5.66	6.08	0.77	0.54
PE-Core-G	400K	0.525	109.4	10.4	5.00	0.69	0.64	0.366	238.4	2.44	4.57	0.81	0.59
+iREPA	400K	0.458	132.0	7.78	5.02	0.70	0.65	0.322	275.4	1.93	4.59	0.80	0.61
CLIP-L	100K	0.790	54.07	25.2	6.65	0.61	0.60	0.605	124.5	9.77	6.59	0.75	0.52
+iREPA	100K	0.657	74.46	17.8	6.33	0.65	0.61	0.489	172.0	5.67	6.33	0.78	0.54
CLIP-L	400K	0.487	117.8	8.97	4.98	0.70	0.65	0.339	258.7	2.15	4.64	0.81	0.59
+iREPA	400K	0.442	130.1	7.42	5.03	0.71	0.65	0.307	271.8	1.96	4.64	0.80	0.61

Table 8: **Additional results with alternative evaluation metrics.** We supply supplementary results under the CMMD metric (Jayasumana et al., 2024). All results are reported using SiT-XL/2 with a batch size of 256 and traditional REPA as the baseline; we adopt the PyTorch implementation of (Paul, 2024) to compute the CMMD metric, and employ the standard reference set from the OpenAI ADM evaluation suite (Dhariwal & Nichol, 2021) as reference images. Across the range of vision encoders, accentuating the transfer of spatial features (iREPA) serves to accelerate convergence under both CMMD and the traditional evaluation metrics (IS, FID, etc.), thereby attesting that the key empirical findings of §4 are robust to the choice of evaluation metric.

M.6 SPATIAL IMPROVEMENTS WITH SAM2

Table 9 and Figure 30 examine the impact of spatial normalization (§4) upon the SAM2-S (46M) encoder features². As Table 9 indicates, we observe that for SAM2, while the spatial normalization layer does improve performance, the improvement can be marginal. The reason is that spatial normalization relies upon the removal of the global component (the mean of the patch tokens) to heighten spatial contrast; since SAM2 already harbors little to no global information (validation accuracy < 24%), spatial normalization improves the spatial contrast only slightly (see Figure 30).

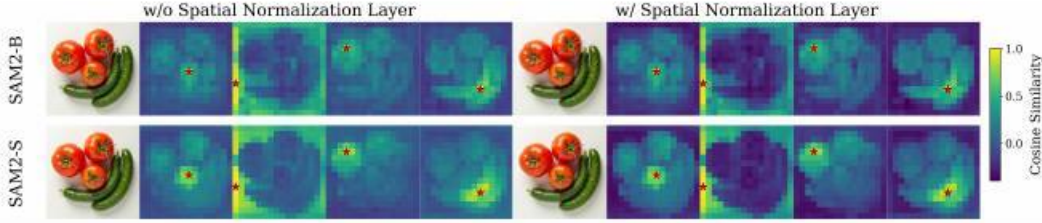


Figure 30: **Impact of spatial normalization on SAM2.** We observe that for SAM2, although the spatial normalization layer does serve to enhance the spatial contrast, the improvement can be marginal. The reason is that spatial normalization (§4) relies upon the removal of the global component (the mean of the patch tokens) to heighten spatial contrast; since SAM2 already harbors little to no global information (validation accuracy < 24%), spatial normalization improves the spatial contrast only slightly.

Vision Encoder	Steps	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
SAM2-S	100K	50.69	25.32	6.52	0.631	0.588
+iREPA	100K	53.14	24.52	6.28	0.629	0.591
SAM2-S	400K	110.77	9.54	4.93	0.699	0.638
+iREPA	400K	114.62	9.10	4.89	0.704	0.640

Table 9: **Impact of spatial improvements on SAM2.** All results are reported with SiT-XL/2, without classifier-free guidance, using SAM2-S (46M) as the vision encoder and traditional REPA as the baseline.

²Note that, following (Bolya et al., 2025), we use the intermediate output of the vision encoder for the SAM2 features rather than the mask logits. As (Bolya et al., 2025) show, although the mask logits yield sharper spatial maps, they are themselves unsuitable as a target representation.

M.7 ADDITIONAL RESULTS WITH FULL-FINETUNING ACCURACY INSTEAD OF LINEAR PROBING ACCURACY

Correlation analysis with full-finetuning accuracy instead of linear probing accuracy. We repeat the correlation analysis of §3 using the validation accuracy obtained after full-finetuning³ in place of linear probing. The results are reported in Fig. 31. All results employ SiT-XL/2 (100K steps) and REPA. For the full-finetuning validation accuracy, in lieu of linear probing, we fully finetune the vision encoder with a learning rate of $5e-5$, a warmup ratio of 0.1, and a total of 30 epochs. Every spatial metric correlates far more strongly with generation performance than does full-finetuning accuracy. Intriguingly, gFID in fact exhibits a weak positive correlation with the validation accuracy after full-finetuning (Pearson $r = 0.317$)—that is, as validation accuracy rises, gFID rises and generation performance deteriorates. This accords with the observations of §2, wherein target representations of higher global semantic performance frequently deliver comparable or worse generation performance under representation alignment (REPA).

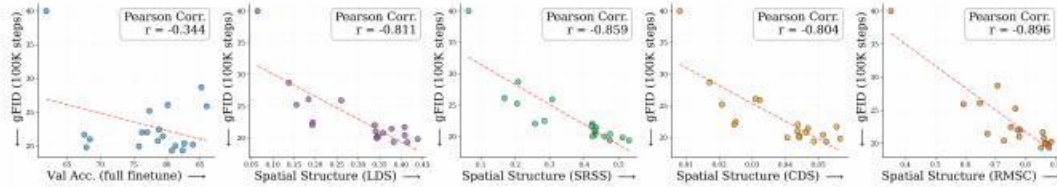


Figure 31: Correlation analysis with full-finetuning accuracy instead of linear probing accuracy. We repeat the correlation analysis of §3 using the validation accuracy after full-finetuning in place of linear probing. All results employ SiT-XL/2 (100K steps) and REPA. Every spatial metric correlates far more strongly with generation performance than does full-finetuning accuracy. Intriguingly, gFID in fact exhibits a weak positive correlation with the validation accuracy after full-finetuning (Pearson $r = 0.317$)—that is, as validation accuracy rises, gFID rises and generation performance deteriorates. This accords with the observations of §2, wherein target representations of higher global semantic performance frequently deliver comparable or worse generation performance under representation alignment (REPA).

³Please note that, although the ultimate findings remain similar, in the context of REPA linear probing may prove more accurate for estimating the global information of “pretrained” vision encoders. This is because REPA regularizes with the “pretrained” encoder features themselves, and finetuning the encoder can itself alter the amount of global information. As in REPA (Yu et al., 2024), we therefore adopt linear probing accuracy as our default measure of global information.

M.8 ADDITIONAL RESULTS ON PIXEL-SPACE DIFFUSION MODELS

Method	#Params	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
PixelFlow (Chen et al., 2025)	459M	24.67	54.33	9.71	-	0.58
PixDDT (Wang et al., 2025b)	434M	36.24	46.37	17.14	-	0.63
PixNerd (Wang et al., 2025a)	458M	43.01	37.49	10.65	-	0.62
DeCo w/o $\mathcal{L}_{\text{FreqFM}}$	426M	46.44	34.12	10.41	-	0.64
DeCo + REPA (Ma et al., 2025b)	426M	48.35	31.35	9.34	-	0.65
JiT (Li & He, 2025a)	459M	29.37	49.06	11.21	0.40	0.62
JiT+iREPA (Ours)	459M	50.72	29.19	9.42	0.51	0.65

Table 10: **Generation performance of pixel-space diffusion models.** Its simplicity notwithstanding, iREPA—when combined with JiT (Li & He, 2025a)—outperforms recently proposed state-of-the-art pixel-space diffusion methods, DeCo (Ma et al., 2025b) among them. All results are reported at 200K training iterations with a batch size of 256 and evaluated using 50-step Euler sampling without classifier-free guidance. Results for DeCo are drawn directly from (Ma et al., 2025b).